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Review article

Toward the mutual routing security in wide area networks: A scoping review of current threats and countermeasures☆

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ABSTRACT

Background: The inter-domain routing security is often based on trust, which, as seen in practice, is an insufficient approach. Due to the deficit of native security controls in the Border Gateway Protocol (BGP), many new routing security measures were proposed to prevent control-plane abuse. They must be implemented in the majority of the Internet's Autonomous Systems.

Objective: This study undertakes a scoping review of the routing security domain to provide the most up-to-date and state-of-the-art broad summary of threat classification, prevention, and mitigation. The authors determine the progress of implementing countermeasures and explain the obstacles and research directions.

Results: This paper covers the current threat landscape and the existing taxonomies of attack vectors on the routing layer. By analyzing different taxonomies, we detected overlapping incident types. Therefore, a unified and consolidated taxonomy is proposed to preserve consistency among different attack types, simultaneously giving a more detailed breakdown of incident classification. This review also contains a comprehensive comparative study of protective measures, including historical, current, and developing techniques. This study includes the efficiency of proactive (prevention) and reactive (mitigation) practices and their caveats. The authors also examine the most promising development plans for new and existing countermeasures.

Conclusion: Global implementation efforts are focused on routing security's safeguard mechanisms based on mutual protection, e.g., the Resource Public Key Infrastructure (RPKI) system. This determinant creates an infinite regress problem known as the chicken or the egg—the primary dilemma. The authors find that the point of critical mass is achieved but that RPKI still faces vital issues.

1. Introduction

Autonomous Systems (ASes) are the primary essential entities in today's Internet. As defined in [1], "an AS is a connected group of one or more IP prefixes run by one or more network operators that have a *single* and *clearly defined* routing policy". Traffic is allowed to flow when a routing policy is configured and accepted by adjacent ASes (i.e., peers). If traffic traverses in an unintended fashion (conflicting with the routing policy), this can be called a *routing anomaly*. Furthermore, if the routing anomaly results in a harmful effect, it should be considered as a *routing incident*.

The most exemplary high-profile routing incident is a sub-prefix origin hijack. As illustrated in Fig. 1, the attacker announces a more specific prefix of the victim's network to the Internet with a changed origin. Assuming that the advertisement can propagate across the Internet, the attacker will cause traffic redirection.

Routing security has been a subject of study for more than two decades. We believe this scoping review benefits the scientific community and policymakers as a comprehensive yet critically created state-of-the-art summary and practitioners who want to secure their networks by implementing the security measures described in this review.

As the source material for the review, we considered any valuable research article in the field of routing security. To enrich the review in bleeding-edge research directions, we benefited from the addition of gray literature and Internet Engineering Task Force (IETF) documents, such as Request For Comments (RFC), non-expired Internet-Drafts (I-Ds), and Best Current Practices (BCPs). However, in order to produce quality content, we chose to exclude works older than ten years unless they talked about legacy solutions and past events.

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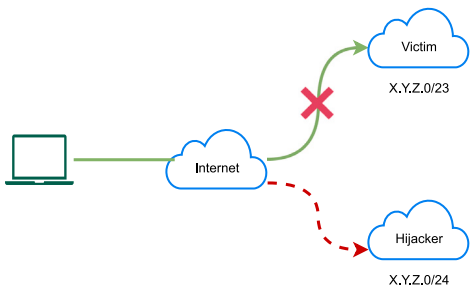


Fig. 1. One of the most typical and effective routing incidents—the more-specific prefix hijack. As a result, traffic is diverted from the victim to the hijacker.

This scoping review was carried out to systematically map the research in the routing security domain. We formulated the following framing questions.

- RQ1: What are the primary causes for routing incidents?
- RQ2: How to address and differentiate incident characteristics?
- RQ3: How to unambiguously specify the type of routing incident?
- RQ4: What protection measures are available and feasible to implement?
- RQ5: How to deal with routing incidents in an efficient and orderly fashion?
- RQ6: What are the most promising research and development directions?

For the threat landscape and taxonomy scope, we reviewed any attack that may be executed due to BGP design shortcomings; therefore, we exclude the vulnerabilities of underlying protocols. We also arbitrarily chose to omit the issue of routing anomaly detection, as we plan to cover it in a future paper.

Regarding protection measures, we focused on practical technologies that improve the currently dominating BGP protocol. As has been shown numerous times already, replacing BGP is not an easy endeavor. The deployment of a control measure that does not compete with BGP has a greater chance and a better expectation of being implemented globally on the Internet. Therefore, we chose to exclude from this review methods and technologies that aim to replace BGP, such as blockchain-based routing, centrally managed Software-Defined Networking (SDN), or Scalability, Control, and Isolation on Next-Generation Networks (SCION). Nevertheless, we briefly discuss these ideas, too.

In summary, this review provides the following contributions:

- Compares two different existing taxonomies to identify overlapping features,
- Introduces a new 5-tuple benchmark for classification of the full spectrum of routing anomalies,
- Indicates simple-yet-accurate key factors in differentiating routing incident characteristics,
- Comprehensively reviews routing incident protection measures from the blue team perspective,
- Provides a knowledge base on handling and mitigating routing incidents,
- Compiles cutting-edge news and outlines potential research directions.

The rest of the paper is organized as follows. First, in Section 2 we discuss relevant related work. Then, in Section 3, a detailed breakdown of routing incident types is presented. Next, Section 4 describes the threat landscape and risk analysis, including incident examples and their classification. The protection and mitigation tactics that reduce the impact of routing attacks are characterized in . If an incident has occurred, the network operator team should deploy one of the mitigation strategies mentioned in Section 6. In Section 7, we summarize the global progression of security measures' implementation and provide a prognosis for future efforts. Finally, Section 8 presents an overview of caveats, open questions, and future development directions, while Section 9 concludes our work.

2. Related work

Over the past several years, significant effort has been put into uncovering new routing vulnerabilities and developing new defense techniques. The urgent need to patch routing vulnerabilities has caused research in the routing security domain to bloom. This can be perceived by the overall publication count and numerous IETF draft proposals. The ongoing "competition" for the leading technology standards and the lack of the universal adoption of them is the driver for emerging new proposals. Although numerous attempts have been made to enumerate and classify attack types and defense mechanisms, the scientific community seems to lack a comprehensive systematic review of the routing security domain. Table 1 summarizes the existing related work.

The excellent survey by Mitseva et al. [2] provides an overview of the countermeasures against BGP attacks. However, it is biased toward theoretical attempts that, unfortunately, have been abandoned. Our review builds on these foundations, providing a more up-to-date and practitioner-focused selection. Researchers have extensively discussed the threat and attack landscape, while ignoring already defined and well-known taxonomies. In the article, there is also a comprehensive review of anomaly detection and localization techniques, summarized in the conclusion because there is no universal method, and most of the existing ones are impractical.

An article by Siddiqui et al. [3] surveys the efforts of the IETF Secure Inter-Domain Routing (SIDR) Working Group (WG) in securing the BGP protocol. It is a highly appreciated summary of the ongoing work to protect the BGP protocol and points out areas that need addressing. However, since its publication, new countermeasure proposals have arisen. The article also refrains from classifying attacks and taxonomy; therefore, our review fills this gap.

Al-Musawi et al. [4] focus on reviewing several BGP anomaly detection techniques according to their approach and capabilities, which is not a feature of this review. The article also provides a classification of BGP anomalies, although shifted from the IETF point of view [5]; thus, it should be noted that it gives a unique perspective. However, no mitigation methods or security solutions are mentioned.

Sermpezis et al. [6] propose a novel detection method and reactive approach to routing incident handling. It is an impressive and comprehensive work on malicious hijack classification, detection, and mitigation, and the results are confirmed by the open-source project ARTEMIS. The spot-on taxonomy is discussed further in our paper. However, this work does not attempt to address the route leak problem.

A survey [7] by Huston, the Chief Scientist at APNIC[1], introduces BGP mechanics, operation principles, and security requirements. There is a general taxonomy of BGP problems, unfortunately, without a detailed drill-down to different types of hijacks or route leaks. The author discusses selected tools for BGP protocol session transport security, most of which are well-known and defined as the industry standard [8]. The survey enlists several ways of protecting BGP message integrity. The author extensively describes the history of BGP security solutions, dating back to 1988. Recent IETF activity is addressed in the article as well. However, newer proposals (such as the OTC attribute or the DO community) are not covered. Huston also refer to some fresh ideas of the BGP transport security, e.g., BGP over TLS, BGP over QUIC, and also redefining RPKI architecture with a push model. The

[1] Asia Pacific Network Information Centre—the regional Internet address registry (RIR) for the Asia-Pacific region, further explained in Section 3.1.

Table 1
Feature comparison of related work.

	Siddiqui et al. [3]	Sriram et al. [5]	Al-Musawi et al. [4]	Mitseva et al. [2]	Sermpezis et al. [6]	Sermpezis et al. [9]	Huston [7]	This paper
General								
Year	2015	2016	2017	2018	2018	2018	2022	2023
Type of work	survey	RFC	review	survey	research	survey	survey	scoping review
Transport security								
Taxonomy	□	□	□	□	□	□	■	⊠
Protection	□	□	□	□	□	□	■	⊠
Origin Hijacks								
Taxonomy	⊠	□	■	■	■	□	□	■
Detection	□	□	■	■	■	⊠	⊠	⊠
Protection	■	□	□	■	□	■	■	■
Mitigation	□	□	□	⊠	■	■	□	■
Path Hijacks								
Taxonomy	⊠	□	■	■	■	□	□	■
Detection	□	□	■	■	■	□	⊠	⊠
Protection	■	□	□	■	□	□	■	■
Mitigation	□	□	□	⊠	■	□	□	■
Route Leaks								
Taxonomy	⊠	■	■	⊠	□	□	□	■
Detection	□	□	■	■	□	□	□	⊠
Protection	■	□	□	■	□	□	⊠	■
Mitigation	□	□	□	⊠	□	□	□	■
Status								
Research challenges	■	□	■	■	□	□	■	■
Operational challenges	■	□	□	⊠	□	□	⊠	■
Adoption issues	■	□	□	⊠	□	■	⊠	■
Deployment progress	■	□	□	□	□	■	□	■

□—no feature; ⊠—partial feature; ■—with feature.

anomaly detection issue is mentioned briefly, while attack mitigation is completely omitted.

Compared to the articles mentioned above, this paper describes and compares the existing routing anomaly taxonomies, while giving new and universal key classification indicators of anomaly characteristics. It further explores protection measures and mitigation techniques – both well-known and newcomers – providing the complete best practice guide on protecting routing and a knowledge base on routing incident handling. Finally, this paper studies the difficulties and obstacles in the routing security patching process, giving a better understanding of the challenges that all new technologies have to face.

3. Routing incidents problem definition

In general, a routing anomaly is an undesirable routing system change that puts the system into a state where at least part of it is not operating any longer, not working the intended way, or not operating at optimal capacity. If a change results in a harmful effect, it should be considered a routing incident.

3.1. BGP primer

This article assumes that the reader is familiar with the fundamental behavior of the BGP protocol. However, we provide a brief introduction of the key concepts, in the belief that they benefit the novice reader for better understanding the routing principles.

The adjacent routers implementing BGP version 4 [10] are called the BGP speakers. In a typical scenario, when a network topology changes, the *Network Layer Reachability Information* (NLRI) payload with *Path Attributes* metrics is exchanged between them in an UPDATE message. The NLRI consists of a prefix and a subnet length; Path Attributes include an AS-Path and an Origin, denoted as an Autonomous System Number (ASN), a unique number allocated to each AS.

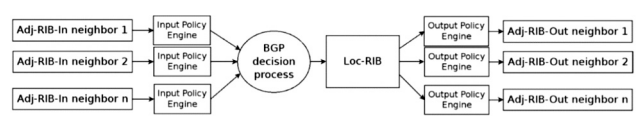


Fig. 2. The BGP protocol process flow [12].

All of the BGP-4 routers implement the best path algorithm, as defined in [11]. This decides which paths will be installed in the routing table and used for traffic forwarding.

However, the decision is a center point in a broader process flow, which is illustrated in Fig. 2 as a block diagram of the flow . BGP routers are usually connected to multiple other routers, and, therefore can receive multiple routes to the same destination prefix. The neighboring routers fill *Adj-RIB-In* tables. After receiving the UPDATE message, the router checks it against *the input filter* to see if the prefix is eligible for further processing. Then, *the decision process* evaluates the best route to each of the received prefixes and installs it in the *Local Routing Information Base* (Loc-RIB). If the Loc-RIB is changed, other neighbors will be notified as long as *the output filters* allow it, through the *Adj-RIB-Out* tables.

Routers use the Longest Prefix Match algorithm for selecting an entry from the routing table. It is expected that multiple prefixes may cover a single IP address. BGP selects the most specific prefix (with the longest subnet mask) for routing purposes. The opposite – the default route – is the shortest prefix of the zero-length subnet mask, which is used as a fallback mechanism.

The Default-Free Zone (DFZ) is a set of all Autonomous Systems that can route packets to any destination without a default route. While it is impossible for a single route to have a complete view of all existing routes, a collection of DFZ routers have a complete BGP routing table,

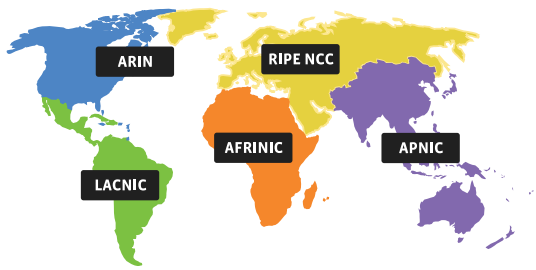


Fig. 3. The RIR map.
Source: IANA [16].

sometimes referred to as the Internet routing table, the global routing table (GRT), or the global BGP table.

Another key concept is *valley-free routing*. The Internet's routing system is constructed on top of tens of thousands of independent autonomous systems. Each is run by one or more network operators, consisting of one or more IP prefixes, and has a single and clearly defined routing policy [1]. There are relationships among ASes, usually backed by a business agreement, but sometimes several ASes can be run by a single operator. Due to the preliminary work of [13,14] defines the relationship roles—provider (p), customer (c), sibling (s), and their behavior, the set of rules known as the *Gao-Rexford condition*. This assumes a hierarchical pattern, "an uphill segment of zero or more c2p or s2s links, followed by zero or one p2p links, followed by a downhill segment of zero or more p2c or s2s links" [15]. The paths that comply with the condition are *valley-free*.

The Internet has strictly defined resource management procedures and hierarchy. The *Internet Assigned Numbers Authority* (IANA) delegates *Internet Number Resources* (INR) such as IP addresses and an *Autonomous System Number* (ASN) pool to *Regional Internet Registries* (RIRs). An RIR is an organization that manages the allocation and registration of an INR within a region of the world (Fig. 3) and delegates resources to its customers, which include *Internet Service Providers* (ISPs) and end-user organizations. Collectively, the five RIRs (RIPE NCC, ARIN, LACNIC, APNIC, AFRINIC) participate in the *Number Resource Organization* (NRO). A *Local Internet Registry* (LIR) is an organization that an RIR has allocated a block of IP addresses to which assigns most parts of this block to its customers.

3.2. Routing incidents' characteristics and taxonomy

The Border Gateway Protocol was proposed in 1989 to replace the Exterior Gateway Protocol. The current version 4 was released in 1994, in the early days of the Internet. Therefore, the designers did not consider protection against accidental errors, malicious data modification, or forging attempts. These BGP vulnerabilities have been the subject of study since the creation of the protocol. A complex threat analysis [17] was carried out in 2006. In summary, vulnerabilities can be divided into four fundamental categories:

- *Lack of native protection of message integrity, confidentiality, and authenticity*. As a result, it is possible to inject bogus routing information and completely disrupt communication. Spoofed actors can impersonate a legitimate BGP speaker. Such vulnerabilities are inherited from the TCP/IP protocol stack characteristics, but they can be easily mitigated with the TCP MD5 signature (Message-Digest algorithm 5) [18] or with the modern TCP-AO (Authentication Option) [19]. The Generalized TTL Security Mechanism (GTSM) [20] can verify the adjacency of the BGP speaker, preventing the remote adversary from communicating. The IP Security (IPsec) tunnel [21] can be established to protect the confidentiality, integrity, and replay protection of routing information exchange with the addition of peer authentication. Some additional remedies are also covered in [8]. As this class of vulnerabilities is more focused on the underlying transport protocol mechanics, we consider it beyond the scope of this paper.
- *No means to verify the legitimacy of the origin in the content of an NLRI message received from legitimate BGP speakers*. This inability could lead to the misorigination of prefixes, which we refer to as a *BGP Origin Hijack*.
- *No means to verify the authenticity of the AS-Path attribute in the content of an NLRI message received from legitimate BGP speakers*. Manipulating the AS-Path can lead to a *BGP Path Hijack*.
- *No means to verify the intended scope of a BGP advertisement in an NLRI message received from legitimate BGP speakers*. The lack of a native mechanism for valley-free policy enforcement leads to a type of incident commonly called a *Route Leak*.

More information on these classes of vulnerabilities are provided in the following subsections.

BGP Prefix Hijacks. A prefix origin hijack event is when an adversary's Autonomous System illegitimately advertises an IP prefix as belonging to another AS. If the hijack is considered successful, the advertisement needs to propagate toward as many other BGP speakers as possible and become accepted to their RIBs. The outcome of this event is that the adversary has modified the DFZ and, therefore, has become the destination AS for the hijacked prefix, which will result in attracting (hijacking) traffic. Protection against this attack is discussed further in Section 5.1.

A path hijack event is denoted when an adversary modifies an AS-Path attribute in order to affect the routing path. Usually, when accomplished on purpose, an adversary wants to attract traffic to their network. This less obvious attack type is harder to execute and simultaneously more challenging to detect. Protection against this attack is very challenging, which is discussed further in Section 5.2.

For the BGP hijack attack class, an extensive taxonomy with three categories is proposed in [6]. Now, we discuss this type of breakdown in short.

The *affected prefix*, that is, the prefix being hijacked, can be the same as the rightfully originating (**Exact Prefix Hijack**) or longer/more specific (**Sub-Prefix Hijack**), while also not being currently advertised (**Squatting Hijack**) by the victim. According to the longest prefix match rule, a sub-prefix hijack advertisement has precedence over the exact prefix hijack. It is worth mentioning that the squatting hijack is easier to execute, as there is no competing advertisement in the BGP decision process.

Furthermore, an AS-Path announced by an adversary is likely to be altered. The researchers of [6] distinguished the **Type-0 Hijack** as when the origin of a route (first AS in the AS-Path) is altered, which is generally referred to as a *BGP Prefix Hijack*. If the AS-Path is altered in a specific way, that attacker is inserted in an *N* position, while the rightful owner is in the first ASN on the AS-Path (the origin), so one can refer to this as a **Type-N Hijack**. For example, if AS1 is the rightful owner and AS2 is a hijacker—the path *{AS2, AS1}* is a Type-1 hijacking, while *{AS2, ASy, AS1}* is a Type-2 hijacking. Additionally, when the AS-Path is not modified, one should distinguish it as a **Type-U Hijack**.

The third category concerns handling data plane traffic when a hijack occurs. Possible actions the adversary can take are **blackholing** when traffic is dropped, using **man-in-the-middle** attack to eavesdrop or manipulate the connection, or **imposture** action when the adversary mimics victim services, and an interaction occurs with legitimate hosts. We discuss the impact in more detail in Section 4.

Route leaks. As defined in [5], a document that aims to provide a common definition for Internet users and the network operator community, a route leak is the propagation of (single or multiple) routing announcements beyond their *intended scope*. The intended scope is defined by the routing policy and is implemented by a set of router

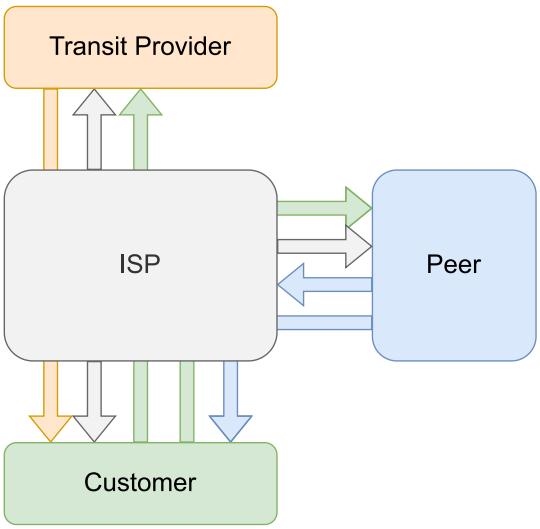


Fig. 4. BGP usual propagation flow (based on [22]). The color of the arrow denotes the origin of the prefix. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

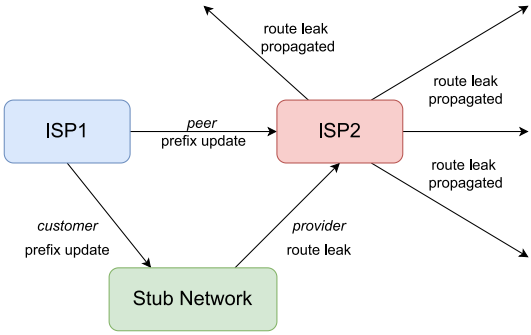


Fig. 5. Route Leak topology (based on [5]).

redistribution filters. It has to comply with the usual BGP propagation flow, the *valley-free routing*, which we discussed in Section 3.1. Therefore, prefix propagation is restricted in terms of both the BGP usual propagation flow presented in Fig. 4, which uses the BGP relationship model proposed in [14,15] and the business relationships between autonomous systems.

Route leaks are typically not malicious and are most often an effect of a configuration mistake. The most common scenario for a prefix leak, illustrated in Fig. 5, is when a multihomed stub AS re-advertises a prefix from one of its transit providers to another. If the second provider has incorrectly configured filters, the leak is not detected, and the route continues to propagate.

However, the IETF suggests [5] a more detailed classification of leaks, which we briefly present.

Type 1: Hairpin Turn with Full Prefix—in the most common scenario, the offending AS is a stub network that leaks a prefix from one transit network to another upstream. A leak is usually successful, as transit networks usually accept prefixes from their customers. Traffic will reach the destination if the stub network is able to process a higher volume of traffic.

Type 2: Lateral ISP-ISP-ISP Leak—when ISPs are peering, they should announce only their own and their customer's prefixes. A Type 2 leak occurs when that rule is violated, e.g., an ISP propagates another ISP's prefix to the third ISP over the lateral peering agreement.

Type 3: Leak of Transit-Provider Prefixes to Peer—this leak type should be distinguished when the offending AS is propagating prefixes learned from the transit provider to a lateral peer network, effectively providing transit service to the peer.

Table 2
Attack classes comparison.

IETF RFC7908	Sermpezis et al.
Type 1–4	Type-N, Exact Prefix
Type 5	Type-0, Exact Prefix
Type 6	Type-0, Sub-Prefix

Type 4: Leak of Peer Prefixes to Transit Provider—in comparison to Type 3, this leak type has the reverse flow direction. Prefixes learned from the peer are wrongly propagated to the upstream.

Type 5: Prefix Re-Origination with Data Path to Legitimate Origin—an offending AS announces the previously learned prefix, but the original AS-Path is discarded. Effectively, the new announcement loses its origin, as the offending AS has re-originated the prefix. Despite the new path propagation, this case implies a condition of reverse data path existence.

Type 6: Accidental Leak of Internal Prefixes and More Specific Prefixes—this type is different from the Type 5, as the prefixes in the leak originate within the offending AS. Type 6 leaks are commonly observed, and typical causes of this problem include a poorly configured export policy, insufficient aggregation, or weak outbound filters. More specific prefixes should not be present in DFZ when a less specific route exists.

3.3. Discussion

In the two presented taxonomies, each tackling a distinct part of the routing incident threat landscape at first glance, some similarities exist. For example, when observed in the routing table, *Type 1–4 route leaks* from IETF's classification [5] are surprisingly indistinguishable from *Type-N, Exact Prefix* hijacks, as classified in [6]. The overlap can be observed for a *Type 5 route leak with a reverse data path* with the similar case of a *Type-0, Exact Prefix* prefix hijack. Again, a *Type 6 route leak* may look similar to a *Type-0, Sub-Prefix* hijack in the routing table, since the offending AS is announcing unregistered routes, but, in contrast to a malicious hijack, the offending AS owns the announced prefixes. For a cumulative summary, see Table 2.

Therefore, observation of the routing table is simply not enough to differentiate between these incidents. It seems that it all boils down to the offending AS's *intent*—the incident's overriding reason. For all the previously mentioned route leaks, it is assumed that the prefix is learned from a neighboring AS. On the contrary, hijacks are usually executed maliciously, serving the offending AS's purpose.

In continuation of this line of thought, various other metrics can be used to classify routing incidents. In addition to the prefix type and AS-Path modification already discussed, we propose the 5-tuple benchmark to describe any routing incident characteristics. We also suggest the sample visualization method in Fig. 6, with three different incidents as examples. The visualization makes it possible to distinguish them at a glance.

intent: identifies a purpose, e.g., an incident may take place from an intentional malicious action (i.e., an attack). If performed on purpose, possible causes include traffic interception for espionage, credential theft, censorship, or Denial-of-Service attack (downtime, outage) on a specific website or company. On the contrary, the erroneous configuration resulting from network operator negligence or human error, sometimes called the *fat finger* error, is without an ultimate agenda.

scale: describes the altered routes count. A targeted incident might include only a single prefix or even a sub-prefix, but sometimes quite a few prefixes may be involved. An extreme case would be a takeover of an entire AS address space.

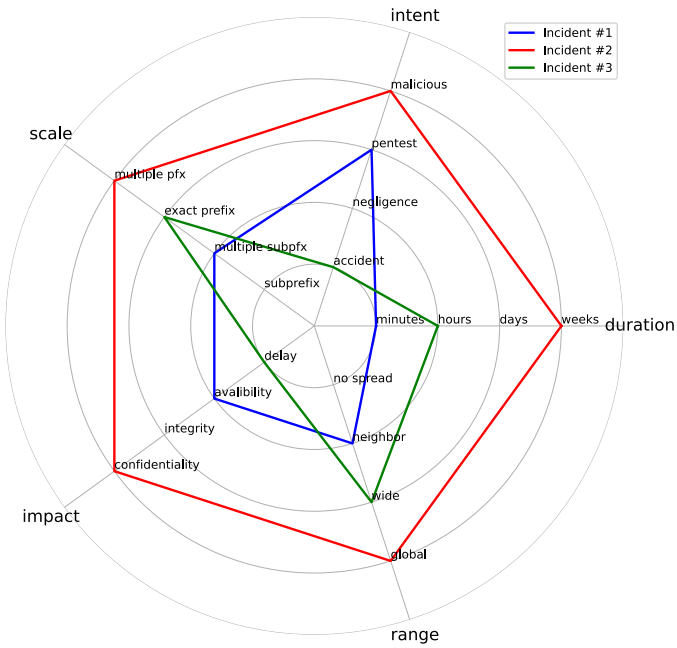


Fig. 6. Benchmarks visualization with hypothetical incidents. The blue line is an incident caused by intentional testing, lasting only several minutes with limited propagation. The red line represents a persistent, malicious, high-scale eavesdropping action with a global range. The green line is accidental, single prefix propagation of limited scale that has caused sub-optimal routing. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

impact: describes the actual damage of the routing incident for the organization, both in the technical and business domains. Usually, there is a correlation between them, e.g., downtime will cause revenue loss, violation of confidentiality may result in credential theft, etc. We dedicate the entirety of Section 4 to a discussion on impact.

duration: identifies how long it takes for the offending prefix to become withdrawn. In normal conditions, legit AS-prefix tuples are long-lasting [23]. Typically, incidents caused by configuration errors that affect a substantial number of networks or users are corrected in a timely manner. However, a short-lived routing anomaly can be not only a misconfiguration, but also an attempt of DNS poisoning or TLS certificate generation abuse. There have been known cases [24] of hijacking defunct ASes for years to gain unused IP addresses for an advertisement fraud operation.

range: describes a scope of a routing event. A local incident is only accepted by a peer AS; on the contrary, some of them propagate globally to the DFZ. This depends on the hijacker network's connectivity, the prefix's length, the implemented protections, etc. A global range attracts more traffic, but sometimes an adversary might want a tailored local range for a stealth operation.

The proposed 5-tuple benchmark is meant to be used alongside existing taxonomies. It expands the existing classifications in terms of specifying "soft" categorical parameters of incident which facilitates further distinction between them. The benchmark also allows us to easily visualize an incident. In addition, it can be further extended, as additional values can be introduced, e.g., new kinds of intent or impact, as needed.

3.4. Malicious activity accompanying routing attacks

This subsection explains several nefarious techniques used in conjunction with exploited routing vulnerabilities. Typically, maliciously performed traffic redirection facilitates further steps of the attack chain.

Domain Name System Cache Poisoning. This remaps the responses of the DNS resolver, causing the redirection of victim traffic. There are multiple ways of achieving such a result, and a BGP hijack is one of them. A short-lived prefix hijack is executed to poison the victim resolver cache. Typically, short-lived prefix hijacks are believed to be configuration errors, as they are quickly corrected [23]. Without having a profound impact on traffic, this type of concern is often ignored, which works in favor of attack stealthiness.

Researchers show [25] that DNS cache poisoning using prefix hijacking is an efficient way to intercept an email, for a password recovery mechanism and account takeover. In the study, verifying the vulnerabilities of selected Internet resource providers (RIRs, domain registrars, infrastructure providers, Certificate Authorities)—over 47% of domains were vulnerable to sub-prefix hijacking. It is worth mentioning that 20% of domains have all nameservers on a single prefix, and 10% have a single nameserver, so when an adversary hijacks one prefix, the entire victim's DNS infrastructure is compromised.

An account takeover can lead to multiple malicious outcomes, but we accentuate the RIR account impersonation. For example, access to the victim's administrator account can be the outcome of (i) transfer of an IPv4 address block for financial benefit and service disruption; (ii) modification of the victim's legal address, contacts, VAT number, and other details; (iii) IRR and RPKI ROA records' manipulation to facilitate further prefix hijacking in a stealth fashion or more service disruptions if desired; and (iv) account termination.

Downgrading of the RPKI-based validation (ROV) could be another target for adversaries through the exploitation of off-path DNS cache poisoning. Such an attack is possible [26] by redirecting the RPKI relying party queries to another publication point. In the absence of up-to-date ROA records, most RPs, by design, fall back to the "pre-RPKI" state. Therefore, the outcome is changed from *invalid* to *unknown*. As a result, this enables the attacker to maliciously announce IP prefixes with RPKI ROAs issued by their rightful owners (that would have the *invalid* state), which are then accepted on the networks that employ RPKI ROV.

The current best practice is to use multiple instances of RP software to provide redundant and resilient infrastructure. However, researchers found [27] that 63% of ASes that operate multiple RPs use a DNS from a single AS, and 43% use only a single DNS resolver. Communication between unprotected DNS resolvers and PP's nameservers can be hijacked by redirecting RPs to a bogus PP or blocking the DNS lookups altogether. To make things worse, only half of DNS resolvers serving RPs were using DNSSEC (which could prevent redirection), of which 60% used vulnerable configuration.

The evolution of the previous attacks was recently shown in [28]. By making the PP unavailable with the usage of (i) spoofing TCP SYN packets or (ii) triggering a nameserver's rate-limit via spoofed DNS queries, an RP cannot refresh its cache, which concludes in expiring the RPKI objects. The optimized attack version named *Stalloris* involves an adversary's controlled hierarchy of publication points. When a victim RP traverses the tree, attackers deliberately *stall* the victim, making it impossible to complete the refresh on time.

Acquisition of bogus TLS certificates. Typically, during the Transport Layer Security (TLS) certificate issuance process, a Certificate Authority (CA) is obligated to check if the party requesting the certificate has control over the domains included in the Certificate Signing Request. This process, known as domain control verification, is crucial to establishing trust. There are several methods available: (i) an HTTP challenge in an HTML page under a specific URL, (ii) a token in a DNS TXT record, (iii) a WHOIS record, and (iv) others. A cybercriminal will be able

to obtain a valid TLS certificate; as they hijack a resource, they need to impersonate the server and pass all checks that the CA is using to verify the remote side [29]. Such an attack was further executed against multiple existing CAs in [30,31] and there is a severe possibility of unlawfully issuing a valid certificate, as 50% of the domains examined were vulnerable, with a success rate of more than 70%.

The presented mechanism reveals a loophole in a CA's verification procedure. Therefore, using a BGP routing attack can exploit such a bug and lead to further target impersonation by the attacker. Researchers suggest two countermeasures for CA: *Multiple Vantage Point Verification* and *Monitoring BGP Route Age*. However, four years after the original paper, there has been little-to-no improvement in the matter [32].

4. Impact of routing incidents

As mentioned in the introduction, when a routing anomaly results in a harmful effect, we consider it as a routing incident. The harmful effect of an incident might have some consequences on real-life networks, services, or businesses. In this section, we analyze the potential impact a routing incident can produce and give examples of past events.

4.1. Impact estimation

Determining the real impact of a routing incident threat is essential to the organizations and their Chief Information Security Officer (CISO). Estimating possible outcomes is paramount as a part of the risk management and planning strategy, as well as during incident response and post-incident review. Unfortunately, each incident type can pose a different risk. However, the most essential element is indicating the business value of services or resources that were using a hijacked prefix and the methods of handling the data plane that the adversary used.

Prefix hijacks. When an autonomous system suffers a hijack, it can affect availability, integrity, and confidentiality. The impact depends mainly on the intent of the incident, which is closely related to the adversary's agenda. However, to break down the technical indicators, let us analyze them using incident-type data plane traffic handling.

Incidents that result in blackholing will have an availability problem for the victim's network and a possible complete outage (a Denial-of-Service case). The effect is similar to that of a data plane DoS attack. However, hijacks are less frequent than data-plane DoS attacks; therefore, fewer networks can defend themselves against such a threat.

When considering a man-in-the-middle attack, the adversary creates a malicious BGP Hijack, focused on acquiring the data and keeping the connection intact. Exemplary cases include *(i)* credential theft, possible within encrypted channels not protected with Perfect Forward Secrecy, and *(ii)* data manipulation, such as altering a DNS response to redirect traffic toward another server.

The most advanced technique is to impersonate the victim's servers. Using a vulnerability in the TLS certificate provisioning procedure, an adversary can create a perfect mirror of a website that will be indistinguishable for most users. An adversary will use this specially designed website to perform credential theft, credit card information gathering, and active eavesdropping. It may also be possible to present fraudulent content, such as a disinformation campaign.

Route leaks. An accidental route leak will cause a redirection to a different path than intended. This redirection may or may not overload some network equipment and links, which can cause performance degradation, as devices or interconnections cannot process or transfer increased traffic. Symptoms include increased packet delay, partial packet loss, or even discarding (blackholing) traffic.

However, with route leaks created on purpose, any Autonomous System on the AS-Path that forwards the traffic can execute any malicious operation described in the previous section.

4.2. Measuring attack impact in the wild

Hijacks sometimes have a severe and lasting impact. While it is relatively easy to monitor the state of routing in the Internet's control plane, thanks to initiatives such as RIPE RIS[2] (Routing Information Service) and the software built on top of it, it is challenging to verify the real impact, since the affected users and companies tend not to disclose details such as financial losses.

However, an attempt has been made to verify the real impact of routing incidents in a survey study [9] of 75 networks, subscribers to the NANOG[3] and RIPE[4] mailing list. Overall, 41% of the operators reported that their organization was a victim of a hijack in the past. "The vast majority (76%) expects the impact of a hijack to last for a long time (a few hours or more), while opinions are divided on whether the hijack will affect a few or many of their services/clients, indicating that there are concerns both for extended (e.g., route leaks) and limited/targeted (e.g., malicious attacks) hijacks. (...) More than 57% of the hijacks lasted more than an hour, while 25% lasted more than a day; around 28% were short-term hijacks lasting a few minutes (14.3%) or seconds (14.3%)".

4.3. Exemplary incidents

MyEtherWallet, a provider of Ethereum cryptocurrency wallets, was attacked by a BGP hijack on April 24, 2018 [33]. The attack was not carried out directly at MyEtherWallet, but instead on Amazon Web Services' DNS infrastructure. In particular, AWS DNS was not compromised, but the AWS Route 53 prefix space was the subject of a hijack. The resulting outcome was USD 365,000 stolen in Ethereum, despite users being warned of a self-signed TLS certificate. The attacker was announcing more-specific prefixes (*sub-prefix hijack*) and running a *man-in-the-middle* proxy for DNS traffic. The cryptocurrency website was then *impersonated*. This incident can be characterized using 5-tuple benchmark as {intent: malicious, scale: sub-prefix, impact: confidentiality, duration: hours, range: global}.

The Telekom Malaysia (AS4788) and Level3 (AS3549) incident was massive *Type 4* route leak. As described in [34,35], more than 176,000 prefixes learned from the peering session in DEC-IX, were re-advertised by AS4788 to the upstream provider AS3549. This leak resulted in a large amount of traffic being directed to AS4788, which also caused a network overload for Level3. Typical symptoms such as packet loss and increased RTT were observed. Therefore, this incident can be described as {intent: accident, scale: multiple prefixes, impact: availability, duration: hours, range: wide}.

Another case among many route leak incidents resulted in the outage of AWS and Amazon services' unavailability. As described by a ThousandEyes researcher in [36], AS33083 announced AWS prefixes to its transit provider AS5580. This announcement was not in the path before, so this was a prefix leak of *Type 4*. The leak affected cloud services and, therefore, multiple companies. Using 5-tuple benchmark, this incident can be denoted as {intent: accident, scale: multiple prefixes, impact: availability, duration: minutes, range: wide}.

In August 2013, a joint operation of the Italian National Military Police, the company Hacking Team (providing spyware services), and the Italian hosting company Aruba S.p.A. orchestrated the *Type-0* hijack [37] in order to restore connection to the malware-infected terminals. The Remote Control System malware was configured to use the specific control server. However, the company providing the server went out of business, so the communication was lost. With

[2] https://ris-live.ripe.net/

[3] https://www.nanog.org/resources/nanog-mailing-list/nanog-mailing-lists/

[4] https://www.ripe.net/participate/mail/ripe-mailing-lists.

Table 3
Feature comparison of different countermeasures.

	IRR filters	Prefix limit	Peerlock	Peerlock-lite	BGPsec	ASPA	Path-End Val.	OTC attribute	DO community	RPKI ROV
Operation mode										
In-band	□	□	□	□	■	□	□	■	■	□
Out-of-band	■	■	■	■	□	■	■	□	□	■
Features										
Good scalability	⊠	⊠	□	■	□	■	■	■	■	■
Low computational overhead	□	■	■	■	□	⊠[CO]	⊠[CO]	■	■	⊠[CO]
Low storage overhead	□	■	■	■	□	□	□	■	■	□
Low bandwidth overhead	■	■	■	■	□	■	■	□	□	■
Unaltered BGP protocol	■	■	■	■	□	■	□	□	⊠	■
Partial deployment										
Incremental depl. supported	■	■	■	■	■	■	■	■	■	■
Beneficial when part. depl.	■	■	⊠[PL]	■	⊠	■	■	■	■	■
Downgrade attack resilient	■	■	■	■	□	■	■	⊠[DG]	⊠[DG]	⊠
Mitigation										
Prefix Hijack Type-0 Exact	⊠[IRR]	□	□	□	□	□	□	□	□	■
Prefix Hijack Type-0 Sub	⊠[IRR]	□	□	□	□	□	□	□	□	■[ML]
Prefix Hijack Type-N Exact	□	□	□	□	■	⊠	⊠	□	□	□
Prefix Hijack Type-N Sub	□	□	□	□	■	⊠	⊠	□	□	□
Route Leak Type 1–4	⊠[IRR]	⊠[PC]	■	■	□	■	■	■	■	□
Route Leak Type 5–6	⊠[IRR]	⊠[PC]	□	□	□	□	□	□	□	■
Status										
Academic or concept paper	□	□	■	■	■	■	■	■	■	□
Standardized	■	■	□	□	■	⊠	□	■	⊠	■
Software implementation	■	■	■	■	⊠	⊠	⊠	⊠	□	■
Real-life deployed	■	■	■	■	□	□	□	□	□	⊠

□—no feature; ⊠—partial feature; ■—with feature.
[CO] Cryptographic operation can be offloaded to a RP software, reducing the load on a router.
[PL] Peerlock is beneficial only when deployed for majority of Tier-1 and Tier-2 networks.
[DG] Downgrade is possible, but not likely. This can be additionally secured with BGPsec.
[ML] The maxLength usage exposes the possibility to a subprefix hijack.
[IRR] Filtering is possible, but depends on data quality.
[PC] A substantial prefix count is required to trigger the failsafe.

no other means to reconfigure a different control server on the infected devices, Hacking Team suggested hijacking the server's IP block, 46.166.163.0/24, which was not advertised at the time, making this a *Squatting Attack*. This controversial, six-day long, illegally carried-out hijack by Aruba with either the cooperation of or by request of the Italian Police was revealed by a 400 GB leak of the Hacking Team's proprietary data and emails [38]. Considering above, this incident characteristics can be expressed as {intent: malicious, scale: exact prefix, impact: integrity, duration: days, range: global}.

5. Protective measures against routing incidents

Preventing routing incidents is a layered solution. To effectively mitigate risk, preparation effort is vital. Implementing preventive controls in the routing control plane effectively avoids the damage that an incident may cause. Eradicating the incident's outcome is a much more complex (see Section 6 for details). Therefore, we strongly suggest deploying some form of defense beforehand.

In the following parts, we present – customary, operational, and battle-tested – security controls, more advanced security measures that aim to become the future standard, and some noteworthy attempts. We discuss their features, strengths, and weaknesses. A summary of safeguards is presented in Table 3. We would like to begin with a short primer on BGP housekeeping and introduce best current practices. Then, each subsection is dedicated to a specific class of routing vulnerability.

Internet Route Registry (IRR) is a system of independent databases that uses the Routing Policy Specification Language (RPSL). There are over 20 independent instances in the global deployment, only a subset of which is public, e.g., RIPE IRR or the Routing Assets Database (RADb). Network operators ought to publish their routing policies and announced prefixes to the IRR and maintain affiliated records, so other ASes can build route filters using that data. Particularly useful for this is a special record type called AS-SET, which defines ASes downstream networks.

However, independent and distributed deployments of IRR results in non-single root-of-authority, overlapping, and inconsistent data, with no precise method of resolving conflicts. On top of that, there is usually little to no validation of the content being put into IRR, which raises questions about the authenticity and genuineness of IRR records. The RPSL lacks a method for describing the intended expiration time of a record. Poor quality data has an impact on filter-building usefulness.

Prefix filtering. When peering with other ASes, network operators have no control over what other ASes announce; therefore, it is critical to filter what routes will be accepted to the RIB. Filtering policy exclude unwanted routes using a set of criteria. The best results for filtering policy are achieved using a positive security model, i.e., default deny and selected permit; however, explicit violations which we discuss next, should be denied.

There are defined, well-known BGP Operations and Security best current practices [8]. While some are not strictly related to BGP Security, focusing instead on reliability, we include only those deemed most essential for a comprehensive, more complete landscape.

Popular housekeeping action filters too-specific prefixes, as a rule of thumb, that are smaller than /24 and /56 for IPV4 and IPv6, respectively. This action by design reduces the routing table's bloatedness but coincidentally helps to stop part of Type 6 route leaks (leaking internal more-specific prefixes). In addition, any network should not advertise Special-Purpose Prefixes (reserved by IANA) and Bogons (Martians – private and reserved addresses defined by RFC1918, RFC5735, and

RFC6598 – and the unallocated IP address space). There is also no place on the Internet for a private ASN or a too-long or empty AS-Path. Each of those mentioned rules applies to both ingress and egress policies.

Excessive BGP AS-Path prepending increases the risk of successful routing incident. Many network operators use prepending as a traffic shaping tool due to the design of the BGP protocol. A router should choose the shorter route, not the prepended route (assuming that the other criteria are equal). Prepending is a common practice on the Internet and, if done correctly, may be beneficial for traffic steering. However, researchers point out [39] that an incorrect usage called *prepend-to-all* has been observed in more than 8% of IPv4 global routing table prefixes. This excessive practice is about advertising the prefix with multiple origin ASN repetition in the AS-Path to each upstream. Such a practice opens a vulnerability: if another AS announces (mistakenly or on purpose) the same prefix with a shorter AS-Path, the traffic is redirected from its usual, elongated path.

5.1. Safeguards against origin protection

An adversary may falsely announce IP prefixes as his own to reroute Internet traffic. In this section, we review available ways of origin verification.

Ingress filtering policy. The IRR provides a means of validating an originating AS number of BGP announcement messages, which is the rightmost ASN in the AS-Path attribute. In the IRR database, a route object defines the association of an IP prefix with an ASN. Therefore, the ingress policy filter must check the match of the announced IP prefix in an NLRI payload against the origin ASN retrieved from the IRR. If the match is unsuccessful, (a) the prefix was announced unlawfully, or (b) the route object in the IRR database is out-of-date.

Ingress filtering is considered the best common practice among network operators. Up to this day, it is the mainline defense against propagating prefix origin hijacks. Depending on the filter, it can block Exact Prefix Type-0, Sub-Prefix Type-0, and Squatting hijacks.

Resource Public Key Infrastructure (RPKI) is a generic system defined to create, maintain, and securely verify the digital attestation of resource ownership. The standard is defined in more than 40 different RFCs, the creation of which began in 2009–2010. The main objective is to sign records associating the IP prefix with a correct ASN. The X.509 standard was utilized to achieve cryptographic attestation, and five *Trust Anchors* (TA) were created. Each of them is operated by one of the RIRs (ARIN, RIPE NCC, APNIC, LACNIC, AfriNIC). These organizations publish the Certification Practice Statement (CPS) [40]—a document that defines the practices used to validate the claims of Internet Number Resources holders unambiguously, allowing them to issue resource attestation confidently. Typically, RIRs maintain the RPKI infrastructure and provide the service to the LIRs as the *hosted RPKI*. On the contrary, it is possible to run a *delegated RPKI* for creating an independent, distributed system.

Route Origin Authorization (ROA) is an object that states a connection between an IP address block (prefix) and a block holder, i.e., an Autonomous System. It provides the means for a router to verify if the BGP speaker is authorized to originate a particular IP prefix.

An ROA is a cryptographically signed triplet consisting of the ASN, the IP prefix, and the maxLength parameter. These data are defined in Cryptographic Message Syntax (CMS) [41]. The RPKI system is based on X.509, so it inherits the certificate expiry time. In practice, most ROAs are valid for 12–24 months. The maxLength parameter specifies the maximum possible length of a subnet mask, creating the possibility for an ISP to cover multiple more-specific IP prefixes under a single ROA. Note that using maxLength opens up the possibility of a more-specific prefixes path hijack.

Route Origin Validation (ROV) is the mechanism defined in [42] that is used to produce a *validity state* of an ROA that can be used as an additional out-of-band routing's control-plane security mechanism.

The *Relying Party* (RP) software implementing ROV downloads the raw ROA repository from the *Publication Points* (PP) indicated in the *Trust Anchor Locators* (TALs), verifies the signature of all RPKI data objects and other checks, and outputs it as the *Validated ROA Payload* (VRP). During the validation process performed by relying parties, checks are applied (as with any cryptographic object) to ensure that the ROAs have not been manipulated, a date has not expired, the chain of trust is valid, etc.

The VRP is fed to a router with the RPKI-Router Protocol (RTR) [43], and then it is checked against the BGP announcements. There are three possible outcomes of this action.

Valid The prefix announced is found in the VRP.

Invalid An announced prefix has a wrong origin ASN, or its prefix length is longer than the maxLength parameter.

NotFound The prefix could not be found in the VRP.

With this knowledge, it is possible to influence the construction of a router's RIB. The routing policy can specify an action that can be taken based on the validity state of an announced prefix. In order to block invalid announcements, prefixes must be rejected. However, during the initial setup, it may be desirable to decrease the preference of the invalid route or tag it with a specified community to assess the impact. However, the mentioned techniques are meant to be temporary, as manipulating communities outside AS based on the ROV state is considered harmful. Any prefixes with the NotFound state should be accepted during the RPKI's incremental enrollment grace period. However, in the final state of RPKI deployment, operators who want a more secure and hardened infrastructure might consider rejecting NotFound prefixes.

5.2. Safeguards against AS-path manipulation

Protecting the integrity of the entire path of BGP route announcements, also known as BGP AS-Path attribute validation, has been a bumpy road. Numerous standards and ideas have been proposed and usually abandoned afterward. To this day, there is no clear candidate for the dominant solution, even with the proposed IETF standard [44]. The existing state of affairs is due to the fact that a solution can be considered viable if it meets numerous features: scalability, low computational cost, no modification to the BGP protocol, and partial deployment support.

In this subsection, we briefly discuss previous proposals for securing AS-Path that did not gain production status but left their mark on the shape of the current solutions. We show how they work and characterize the advantages and disadvantages of each method.

Secure BGP (S-BGP). The specification was proposed in 2000 in [45], adding digital signatures and an attestation to both the AS-Path and prefixes inside the UPDATE message, which are securely transferred in the IPSec tunnel. This solution had significant drawbacks: more powerful hardware was needed to validate digital signatures, and a modification of the BGP protocol was needed. Furthermore, as proven in [46, Section 3.1], the partial deployment was vulnerable to a downgrade attack if it did not follow the idealistic "security 1st model".

Inter-Domain Route Validation (IRV). This proposal from 2002 [47] used an out-of-band validation method, so it did not require modifying the BGP protocol. Contrary to previous ideas, the IRV provided independent databases decoupled from BGP that could be queried on prefix origin or AS-Path verification. However, no practical examples or guidelines were given.

Pretty Secure BGP (PS-BGP). Proposed in 2005 [48], this protocol was able to ensure the authenticity of the BGP speaker, the AS number, the verification of the prefix origin, and also the AS-Path. In addition, it provided data integrity for BGP updates. However, its main drawback

was using IRR, with all of its flaws, to store authentication data, which could not be reliably verified. PS-BGP also provided partial AS-Path validation, which was supposed to simplify partial deployment but is considered unsafe.

Secure Origin BGP (soBGP). This was specified in an IETF draft [49] dated 2006, which is currently expired. The technology was primarily focused on securing origin validation; however, it added the assurance of path plausibility. Contrary to Secure BGP, it used the Web of Trust, which could lead to a more feasible deployment, but at the cost of weaker security. Paths that were registered are considered more trustworthy. A rating function could be used to prefer those paths and reject untrusted routes. However, it was unable to filter the route leaks.

BGPsec. This idea emerged in 2011 and was finally standardized in [44,50] in 2017. Although it provided powerful security features, such as protection against false AS-Path advertisements, it failed to stop route leaks because the BGP policy was not verified. It is also known that BGPsec was vulnerable to withdrawal replay attacks, withdrawal route starvation, and, most importantly, downgrade attacks because of the permission of the partial deployment scenario. Among the implementing software is BGP-SRx [51] by the National Institute of Standards and Technology (NIST) and the BGPsec-enabled Bird Routing Daemon.[5] The BGPsec protocol shared resource intensity with its predecessor, S-BGP. The increased hardware requirement was considered one of the showstoppers. However, there were attempts [52] that aimed to reduce memory footprint. Considering the mentioned flaws in the design of BGPsec, it is unlikely that a rapid deployment will occur.

5.3. Safeguards against Route Leak

Opposite to the AS-Path manipulation countermeasures, route leaks protections tend to focus on network relationship schemes. Their primary goal is to enforce the correct flow of prefix reachability information (routes) across multiple Autonomous Systems according to rules of *valley free routing*. It is worth mentioning that the IETF Secure Inter-Domain Routing Working Group (SIDR WG) acknowledges route leaks as a persisting, residual routing security vulnerability [53]. Securing AS policies is considered out of the scope of BGP semantics. However, this requirement is perceived as a requirement for the IETF Global Routing Operations Working Group (GROW WG) [54].

Egress filtering policy. Controlling the prefixes advertised to other ASes is the first step in preventing a route leak altogether. The most common practice is to enforce the routing policy on the peering (edge) routers based on the community tag placed on inbound traffic. In addition, any internal traffic engineering, such as sinkhole or black hole routes, should also be discarded.

The rule of valley-free routing is that any network must send only locally originated prefixes and its customer's prefixes (the AS-SET) toward the upstream or peer. Contrarily, the upstream transit provider should send the internet routing table (the Default Free Zone) or a subset of it, depending on the contract.

Ingress filtering policy. Using the IRR data, it is possible to create ingress filters capable of preventing the acceptance of a leaked route into the local routing table, e.g., a customer-facing policy should allow only prefixes from its AS-Cone defines as IRR's AS-SET. This method is effective against Type 1–6 route leaks; however, it is only as good as the quality of the IRR data, i.e., it is affected by the IRR's poor condition and inconsistency. In addition, the proper validation capability decreases furthest from an incident's origin, i.e., the transit operator is most likely unable to stop a leak incident, which is taking place a few AS hops away. With an increased AS cone, the prefix count grows exponentially, making the generated prefix filter list infeasible to implement even on high-quality equipment.

Prefix count maximum limit. In a typical scenario [8], a roughly constant number of prefixes is received in a BGP session from a downstream peer. If a failsafe is triggered, it is possible to configure a restriction mechanism that will limit the prefix count or even shut down the session altogether. This protection is effective against massive leaks but will not detect low-profile targeted attacks. This mechanism may also cover the protection of the router against memory depletion. The downside is that it may cause false positives unless the prefix count is updated regularly.

Default route propagation behavior. The standard RFC8212 [55] defined in 2017 updates the default behavior of the BGP router across different software implementations. All routes are neither imported nor exported to other BGP speakers, unless explicitly configured to prevent accidental leaks in fresh installations.

Peerlock, described by the company NTT in 2016 [56], is a rulebook for leak filtering using an out-of-band policy exchange channel. The *protected AS* specifies their authorized upstreams to the *protector AS*, establishing an ASes relationship. With this knowledge, the protector AS will reject all BGP NLRI messages containing the protected ASN in AS-Path, excluding those received from the protected AS and from the authorized upstream with the protected AS adjacent. Note that such a scheme is effective only against Type 1–4 leaks.

As pointed out in [57], Peerlock is lightweight. Compared to RPKI, it does not require any trust or cryptographic infrastructure. Independence from IRR relieves it from the previously described legacy burden. However, the setup might be labor-intensive, as it requires a manual routing policy configuration, with a reconciliation that is based on non-standard AS-to-AS communication.

Peerlock-lite, also known as a *Tier-1 filter*, or a *big networks filter*, is a filtering mechanism based on an empirical presumption, which happens to be a valid constraint within the AS relationship model [15]. Tier-1 transit providers have no provider by definition; therefore, Tier-1 ASNs should not appear twice in the AS-Path [58]. An undoubted advantage of the Peerlock-lite deployment is the ease of implementation and low maintenance effort.

In experimental measurements, it was proven [57] that implementing Peerlock for all Tier-1 and Peerlock-lite in the most sizeable Tier-2 ISPs (still less than 1% of networks) significantly decreases the number of leaked routes. If Tier-2 is omitted, the highly connected peering network of a Tier-2 ISP bypasses Tier-1 Peerlock, allowing a secondary channel for leak propagation. Therefore, implementing Peerlock-lite by Tier-2 ISPs is highly beneficial for the Internet. Interestingly, the combination of Peerlock and Peerlock-lite filtering by Tier-2 ISPs adds little value over Peerlock-lite alone.

Autonomous System Provider Authorization. This [59] is the system for AS-Path attribute verification, primarily intended to solve route leak issues, although it is able to partially protect against AS-Path malicious modification. Autonomous System Provider Authorization (ASPA) uses digitally signed objects that connect a Set of Provider ASes (SPAS) to a Customer AS (CAS) number for the specified Address Family (AFI). In other words, any network (CAS) can define its upstream providers (SPAS). Any ASPA object is stored in the RPKI repository, similar to that of the ROAs. This design decision greatly facilitates ASes already implementing Origin Validation with ROAs. Exactly opposite to IRR's AS-Set, where the provider specifies the downstream networks (aka AS-Cone or AS-Set), it is in the customer's best interest to configure and maintain up-to-date SPAS in the ASPA record. The architecture of a sample network with ASPA object definitions, illustrating the allowed and violating paths, can be found in Fig. 7.

This object database allows other ASes to perform the verification defined in [60]. There are two parts to that procedure. First, the

[5] http://www.securerouting.net/tools/bird/

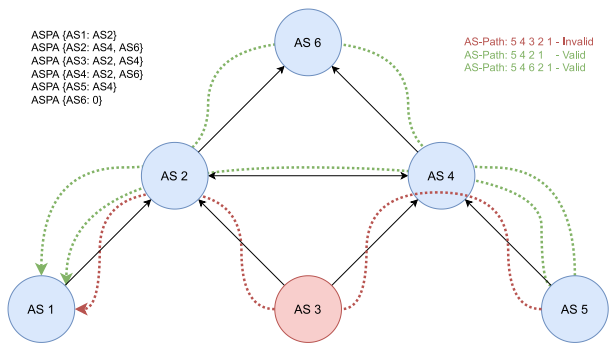


Fig. 7. Exemplary route leak violating "valley free routing" (red arrow), which can be detected and mitigated by ASPA. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

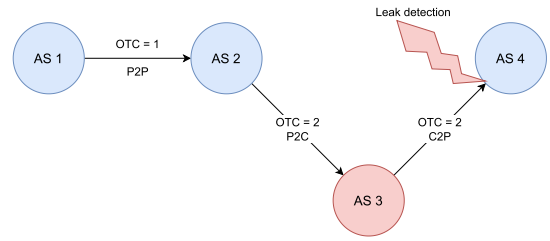


Fig. 8. Route leak detection with OTC attribute (AS3 does not participate in leak protection and starts a leak).

upstream path is verified when a BGP UPDATE is received from a customer or peer. Each AS pair in AS-Path is checked against the ASPA objects retrieved from the repository. When the route is received from a provider, the downstream path is checked. Typically, the entire path has the downstream fragment following the upstream fragment. The opposite order would violate the valley-free routing.

Both steps require checking each AS pair (i.e., {AS1: AS2}) in the AS-Path against the ASPA database for the specified AFI. The process checks if AS2 is adjacent (found among SPAS) to the selected CAS (AS1) and returns one of the following responses.

- Valid: The AS2 is a valid provider for the AS1.
- Invalid: There is no such pair in the ASPAs, so AS2 is not upstream of AS1.
- NotFound: The ASPA for AS1 has not been defined.

The ASPA maintains the right balance between being lightweight in terms of computational resources and being able to detect the majority of route leaks and AS-Path manipulations. Although incremental deployment is provided, there is no risk of downgrading. However, while opting for solving real-world problems, the main caveat is that ASPA replaces path validation with path plausibility. Therefore, it cannot detect every AS-Path forgery or manipulation unless full ASPA objects' publication coverage is obtained.

At the moment of writing,[6] ASPA validation implementation exists for the NIST BGP-SRx version 6 software router. The draft was designed in 2018 and awaits final standardization.

Route Leak Prevention and Detection Using Roles in UPDATE and OPEN Messages. Formerly known as *BGP OPEN policy* [61], this method formally assigns a specific role to each party in a peering relationship to achieve an appropriate configuration and good filtering rules on both sides. The following roles are specified in the context of BGP route propagation constraints.

- Provider: The sender is a transit provider of the neighbor.
- Customer: The sender is a customer of the neighbor.
- Route Server (RS): The sender is the route server at an Internet eXchange Point (IXP).
- RS–client: The sender is the client of an RS at an IXP.
- Peer: The sender and neighbors are lateral peers.

The mechanism does not automatically negotiate the roles between ASes, so, by design, it is required to be input manually firsthand. To achieve compliance, both parties must form one of the pairs in the valid pair set: {Provider, Customer}, {Customer, Provider}, {RS, RS–Client}, {RS–Client, RS}, or {Peer, Peer}. An empty role is also supported, as required by the incremental deployment scenario. However, if the advertised roles do not form one of the allowed sets, the connection must be rejected. The formed relationship enforces proper route exchange and propagation according to the rules of the intended scope.

Moreover, to further reduce route leaks in partial deployment, a new transitive attribute is introduced—*Only To Customer (OTC)*. This allows for the detection and mitigation of route leaks, even multiple hops away. The mechanism works by marking all routes sent to *Customers*, *RS–Clients*, and *Peers* with the OTC attribute equal to the local ASN. If a remote AS detects such a route on a BGP session from a *Customer* or *RS–Client*, it must be rejected. When the route is received from a *Peer*, the OTC attribute must match the remote ASN. The architecture of a sample network illustrating the usage of the OTC attribute is shown in Fig. 8.

The idea was designed in 2016 and finally standardized in 2022. At the moment of writing,[7] RFC9234 implementation exists for BIRD 2.0.11, FRR 8.4, and OpenBGPD 7.5 software routers.

Down Only community. Ref. [62] defines a new well-known BGP Large Community as a signaling measure for route leak prevention, detection, and mitigation. The algorithm is identical to the one defined in the RFC9234—BGP OPEN policy.

On the contrary, this uses the community, not the attribute. Therefore, this does not require any modification in the BGP software, which, in the past, was known to be a bit cumbersome. This simplifies the deployment and adoption. However, it is not guaranteed that the DO community would not be stripped on ingress or egress.

Path-End Validation [63] is an extension of the RPKI system that enables the verification of the legitimacy of the AS-Path last hop. Network operators define valid AS neighboring relations by storing AS adjacency records in the RPKI system. Then, in the background, an agent application retrieves those records and verifies its cryptographic signature to build a relations database. Finally, the agent connects to a BGP router to deploy path filters or produce a separate filter list, to be manually implemented by the network operator.

Researchers claim that the path-end validation, compared to BGPsec, provides substantial security improvement, even in partial deployment. At the same time, it does not require changing the BGP protocol similar to RPKI, and there is no computational overhead for a router.

Recently, the idea was reproduced [64] in a simulated environment. The conclusions were upheld, "finding an improvement from path-end validation which, while not as dramatic as that claimed by the original authors, supports the idea that path-end validation can significantly decrease an attacker's success rate even in minimal deployment".

In our opinion, even though the idea is exciting and we cannot argue with the results, we see strong similarities to ASPA, which is on the final leap to becoming the IETF standard.

[6] As of 2023-01-15.

[7] As of 2023-01-15.

5.4. Discussion

The previous and historical attempts described in Section 5.2 show that achieving a proper, cryptographically verifiable end-to-end signed AS-Path validation is an unachievable endeavor. Since obstacles such as a flawed design, resource intensity, a need for modification of the BGP protocol, and downgrade attacks on partial deployment were encountered, the need for a lightweight solution with no significant changes to the BGP protocol and incremental deployment support has arisen.

There have been many attempts to solve the route leak problem. To date, we have seen only halfway solutions, some addressing only a part of the problem and some being non-standardized. Adequate route leak protection is starting to look like a multilayered solution with complementary mechanisms. Solving this problem requires adopting at least one of the upcoming standards by the most well-connected Internet Autonomous Systems. Please see Table 3 for the mechanism comparison.

The easily deployable Peerlock and Peerlock-lite are tempting, low-hanging fruit but are not the best long-term solution. Their mission is valid; however, drawbacks such as nonscalability, arbitrable AS selection, no standardization, a lack of certainty of malicious leak mitigation, and, most importantly, no active development make them unfit to be a future standard.

The Only-To-Customer BGP attribute and the Down Only Community emerging standards are trying to achieve similar goals—a defined way of detecting far-originated route leaks, with support in the partial deployment. While both are in-band methods, they tackle the problem differently; therefore, they have different pros and cons. The transient OTC attribute is more likely to be unaltered through non-participating BGP routers. Still, adoption requires changes in the protocol, implementation, and software upgrade effort that could delay deployment. On the contrary, defining a new Large Community does not require any modification to the protocol. However, it is technically possible that some networks can remove the community.

It is worth mentioning that both OTC and DO are not cryptographically protected. Therefore, signaling can be maliciously manipulated. It is possible to intentionally leak a route in an undetected manner (false negative) or cause misdetection (false positive). However, this can be fixed in the future by deploying BGPsec. The authors argue that this attack vector is a small price for an effective solution capable of eliminating most accidental route leaks.

When it became apparent that the most prominent contender for AS-Path validation, BGPsec, had many drawbacks, its deployment was impractical and complex, and the need for AS-Path protection remained. ASPA aspires to fill this gap with the goal of stopping the propagation of crafted or invalid paths. It has fine features, such as a lightweight operation that does not require BGP protocol modification, and it provides benefits to early adopters; while simultaneously, it prevents route leaks. However, compared to the OTC attribute, the design is way more complicated, and the setup is also more cumbersome due to its out-of-band nature.

There are also other techniques to mitigate path diversion. During a path manipulation attempt or a route leak, the path is usually longer than the original, thus working against the attacker. Increased connectivity with vital peers decreases the likelihood of successful path hijacking. Engineering a direct peering, or even redundant uplinks with different transit providers, may contribute to the shortening of the AS-Path.

When peering in an Internet eXchange Point (IXP) directly or peering through a route server, there is no need for path validation because the AS-Path length is only one hop (most route servers do not add themselves to the path). Verification of the route origin is sufficient in those cases.

6. Routing incident response

In the case of a successful attack, the victim must take actions to limit the incident's impact. This section provides a juxtaposition of the limited but possible countermeasures a victim AS can undertake, with their pros and cons.

In reality, mitigating a routing incident is a complex and layered task [65]. There is no simple, universal solution because BGP incidents occur outside the victim's network and outside its jurisdiction. This reality implies that crucial changes must be applied outside the victim's organization. Thus, efficient mitigation requires cooperation between the victim's network operator team and upstream Network Operations Center (NOC) or Security Operations Center (SOC) teams.

Prefix deaggregation. In conformance with the BGP protocol design algorithm, it is possible to attract back hijacked traffic by announcing more specific prefixes of the hijacked prefix. This technique is considered a temporary fix or containment solution, as it does not solve the root cause of the problem, though it is effective in most cases. This method's main limitation is its inability to use prefixes longer than /24 because they are usually filtered by Tier-1 and Tier-2 transit operators. Note that the adversary may also announce a more specific route, rendering the solution inoperable.

Shorter AS-Path. Another possibility of attracting back hijacked traffic is again linked to the BGP route selection core algorithm. Supposing that the victim announces the same prefix as the adversary, this time by using other ISPs or deleting unnecessary AS-Path prepends, results in a shorter AS-Path; then, other Autonomous Systems should prefer the victim's legitimate path. It is also worth noting that this method is useless against *Sub-Prefix Hijack*.

Contacting the offending Autonomous System. To restore balance to the routing system, the offending AS has to stop announcing (withdrawing) colliding prefixes. Achieving this may be a challenging undertaking, especially if the prefix is hijacked on purpose. Network operators' contacts are available in the WHOIS database, under the `abuse-c` and `tech-c` organization's keys.

Contacting other Autonomous Systems. If the direct contact with the offending AS fails, it is still possible to limit the *range* and *impact* of the incident by contacting the upstream network (transit provider) of the offending AS. The victim can request the transit provider to filter or drop the false advertisement.

Contacting authorities. The last resort would be to contact the appropriate cyber-crime law enforcement agency (LEA), CERT, or CSIRT for assistance. Computer Security Incident Response Teams possess specialized knowledge of attacks and threats as well as mitigation and resolution strategies. Sometimes local laws may require reporting an incident to the appropriate LEA.

7. Current state of affairs and future work

Retrofitting the security of the routing system is a long, complicated, and still-incomplete endeavor [22]. The RFC for RPKI Route Validation was published in 2010. At first, it was uncertain if RPKI would take off as a practical remedy to routing problems [66], though it is finally gaining some traction due to the culmination of several factors. Institutions such as RIRs and MANRS[8] have been promoting it to network engineers for many years. Multiple *RPKI Deployathon* events focused on implementing RPKI in a live environment, as well as regular RPKI training sessions, have been held. In parallel, increasing impatience with prefix hijacks has resulted in a greater demand for practical routing security countermeasures. Finally, the combination of the most

[8] Mutually Agreed Norms for Routing Security—initiative on securing the Internet routing.

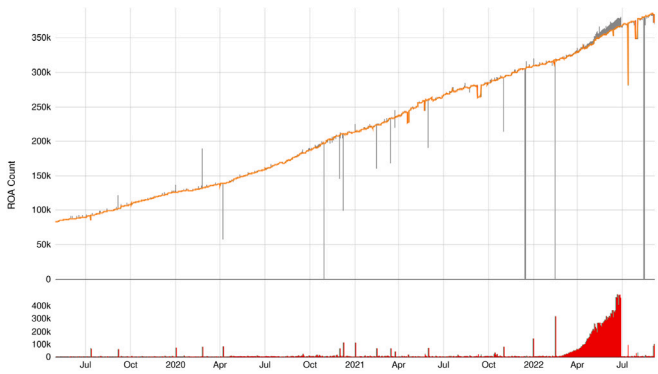


Fig. 9. Total count of ROAs over time. Since the beginning of deployment, the upward trend continues.
Source: [67].

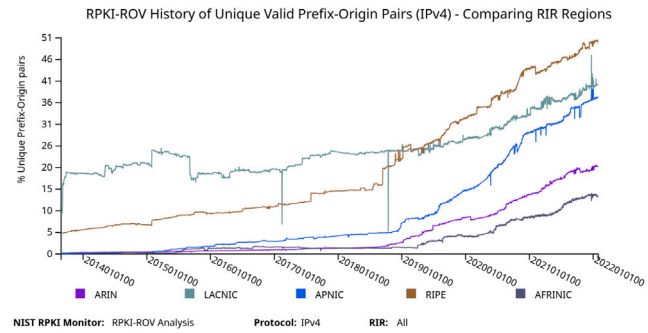


Fig. 10. A percentage of the total count of uniquely announced prefix–origin pairs with prefixes in the corresponding RIR. Around the beginning of 2019, the trend line begins accelerating.
Source: [69].

significant content providers signing their ROAs and the leading transit providers starting an ROV sets an example for the community that defined the tipping point.

Routing security, in general, is challenging to measure, as there are plenty of measurability indicators. Although it is easy to enumerate and count routing incidents to estimate routing health, the implemented routing policy and other countermeasures among autonomous systems are not publicly available. That is why, in Section 7.1, we provide angles on the current state of the routing security controls deployment based on the publicly available statistical data. Next, in Section 7.2, we review the existing questionnaire-based surveys available in the literature. Finally, Section 7.3 focuses on the possible, in our opinion, causes of routing security controls' slow adoption.

7.1. Data[9]

Since the IRR data and the RPKI ROA registry are public, it is relatively easy to estimate the progress of security control implementation. The MANRS Observatory reports that 88.4% of entire Internet prefixes are registered in the IRR. For RPKI, it is 37.4%. However, in the past decade, these numbers appear to be constantly growing, as illustrated in Fig. 9.

However, measuring ROV and IRR filtering is more challenging. A detailed description of the methodologies is beyond the scope of this paper; interested readers can refer to [68]. However, there are some quality measurement data available for the RPKI ROV, such as the NIST RPKI Monitor (see Fig. 10) and the MANRS Observatory, with a current IRR validation rate of 80.5% and an RPKI ROV of 25.3%.

Noteworthy, there is also an RPKI ROV measurement method based on NetFlow [70] (network flow-based telemetry). The idea behind this was to verify the impact of dropping invalid prefixes. The described method checks the volume of traffic in a production network behind the prefixes with respect to their ROA/ROV state. The famous quotation "Not everyone needs to do RPKI" [71] is based on the fact that the Internet has a hierarchical structure. With this in mind, the statement "only 20 companies or so need to do Origin Validation for there to be big benefits" sheds new light on the progress of RPKI ROV deployment. Recently, this idea was examined in [72] for the customer dataset of the network analytical and observability company Kentik. At the time, only 34.1% of IPv4 ROAs were valid, but these valid destination prefixes generated 56.4% of the total amount of traffic. This clearly shows a disproportion, which explains that not every prefix is equal. Note, however, that the presented analysis is of Kentik's customer set, mainly located in the US, which may bias the results.

Another example was presented in the previously cited [72]. According to Madory, the rapid deployment of an RPKI ROA in Poland (see Fig. 11(a)) led to an increase in Kentik's customer traffic being marked as valid (see Fig. 11(c)). Again, the increased value is biased based on Kentik's customer set. However, we determined that the network responsible for the change was AS5617 (see Fig. 11(b)), a Polish Tier-2 ISP. The conclusion is that the actions of a singular Tier-1 or Tier-2 network can have a global impact.

7.2. Existing questionnaire-based surveys

In this subsection, we characterize the current state of inter-domain security backed by existing questionnaire-based surveys. This data, gathered from technical communities, gives us a broader overview of the use of security controls. It also reveals more details on the types of security measures deployed or the reasons behind refusing to use them.

Network operators from NANOG and RIPE NCC communities

This survey [9] was conducted in 2018 among 75 participants—subscribers to the NANOG and RIPE mailing list. The survey was done anonymously and consisted of 21 questions evaluating awareness, implemented defenses, and past incidents.

Most network operators (71%) responded that they had not implemented RPKI as a proactive defense mechanism in their networks. Very few (12%) use the full functionality of RPKI (Route Origin Authorization and Route Origin Validation). An ROA is used by 15% of these networks. However, 60% responders utilize other than RPKI defense mechanism. Fig. 12 presents the visualization of this study. This survey also discusses the routing incident's impact on participants, which we covered earlier in Section 4.2.

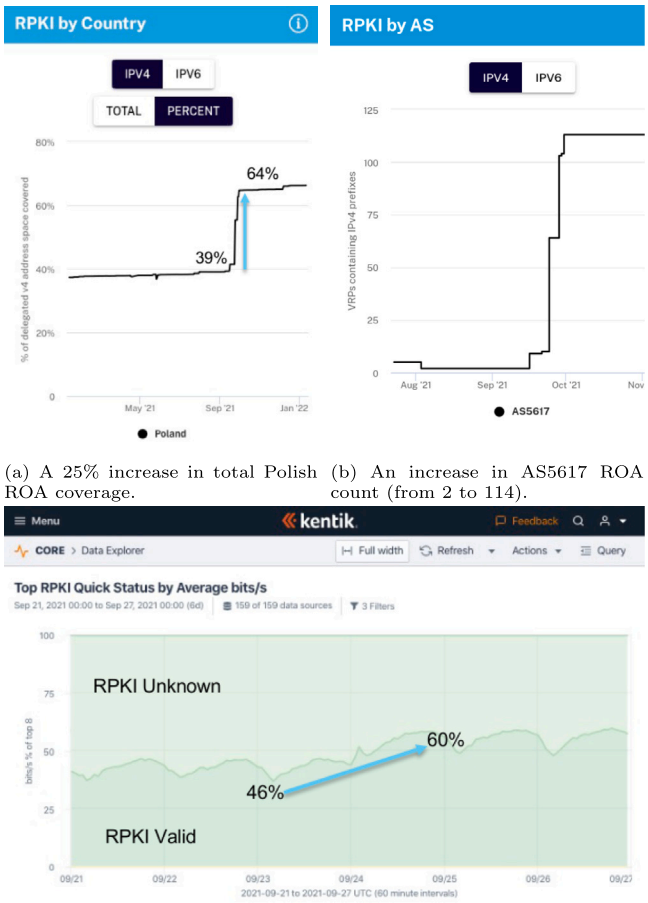
Survey on MANRS members

In 2022, the US Federal Communications Commission (FCC) submitted an inquiry on the vulnerabilities that threaten BGP, security controls, standards, and efforts being made to protect protocol security [73]. One of the answering institutions was the Internet Society.[10] This report [74] identifies key points and objectives for securing the routing. In our view, the most important ones are stimulating the deployment of the solutions mentioned in previous sections, publishing BCPs to improve the adoption of routing security, creating positive incentives, and focusing on high-risk sectors such as critical infrastructure.

A survey was conducted among the MANRS participants to answer the questions in more detail. Even though the survey is biased toward routing security-conscious network administrators and does not

[9] as of 2022-05-22.

[10] https://www.internetsociety.org/



(a) A 25% increase in total Polish ROA coverage.

(b) An increase in AS5617 ROA count (from 2 to 114).

(c) A 14% increase in traffic matching Valid prefixes.

Fig. 11. A correlation between ROAs and valid traffic volume in Polish address space (source: (a)(c)—[72], (b)—RIPE Stat).

represent the whole Internet ecosystem (the sample is relatively small—84 responses, 10% of MANRS' total participants), we find the results quite thought-provoking, providing a unique angle on routing security controls deployment among the most aware network group.

While the world's RPKI ROV adoption is measured to be around 35% or 20% (depending on the source), the survey results show that 67% of participants are using filtering based on RPKI. This number is almost equal to the percentage of IRR filtering, which we hope is on a downtrend. For the main reasons for not creating ROAs, participants point out the complicated process (33%) and the possible failure of an invalid ROA (27%). Peerlocking is used only by 30% of participants, but, more interestingly, almost 18% of them answered "unsure", which is to be understood as a lack of awareness. Of those who do not use it, the main reason (53%) is not fully understanding the risks behind it. Fig. 13 presents the utilization of countermeasures among MANRS participants.

Regarding the BGPsec protocol, there are multiple reasons for the lack of widespread adoption indicated in the report. Similarly, as mentioned earlier, the main ones are the performance problems, the possibility of a downgrade attack, and the lack of route leak protection. Moreover, there is observable disinterest in deploying BGPsec—more than three-quarters of participants did not ask their equipment vendor about BGPsec implementation. It is also worth noting that to this day, there has been no commercial implementation supporting BGPsec.

7.3. Adoption issues

Is routing security a chicken and egg situation? A singular AS does not benefit directly from investing and deploying routing security measures, as the security gain is distributed among all the networks on the Internet. A decades-long but urgent need for security upgrades is a collective action problem—the more network operators cooperate, the higher the number of incident reductions and the lower impact can be expected.

Lack of awareness

Since the RPKI launch in 2010, there have been countless campaigns, lectures, and actions to popularize and encourage deployment among operators. The topic has been covered numerous times [71,72,75,76] during lectures at Information Technology (IT) and Information and Communications Technology (ICT) conferences. The distinctive initiative is the MANRS program for ISPs. By introducing generally agreed norms, MANRS has taken a big step in reducing the most critical routing threats [77]. Overall, given the several years of dissemination campaigns, lack of awareness should be excluded from deployment progress obstacles.

Little security benefit

Researchers who were surveyed [9] network operators from NANOG and RIPE NCC communities pointed out that "deployment lags mainly due to RPKI's limited adoption and little security benefits, but also due to the increased CAPEX and OPEX costs and increased complexity and processing overhead associated with the protocol mechanisms. Therefore, about 60% of the operators resort to other mechanisms and practical defenses to protect their networks against BGP hijacks". For a detailed breakdown of the reasons for adoption slowness, see Fig. 12(b).

In fact, RPKI's minimal security benefit and lack of broad adoption were valid concerns in 2018. As mentioned in the introduction, the mutual interaction of this security measure may lead to a chicken and egg situation. The third reason is investment and infrastructure running costs. They are usually expected with every new technology onboarding. However, we must disagree, as the delegated RPKI ROA is free-to-use, and there is plenty of different open-source relying party software.

Legal issues

Widespread RPKI adoption may be limited due to the specific status of ARIN's terms and conditions for the RPKI repository. Third parties who wanted to validate the BGP routes had to agree to the *Relying Party Agreement*. As a result, most RP-validating software does not have ARIN TAL embedded by default.

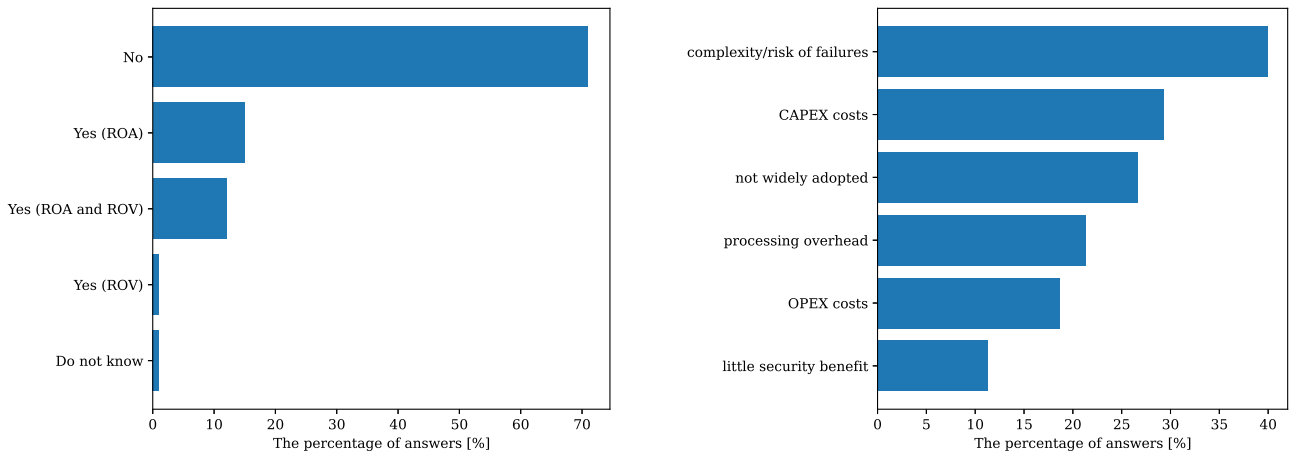
This RPA directly affected ARIN members in North America—Route Origin Validation within ARIN ASes has been significantly lower. In 2018, almost 80% of the ASes involved in ROV omitted ARIN TAL, as demonstrated in [78].

At the beginning of 2019, the gap related to ARIN's ROA only grew. This state of affairs was criticized in [79], and researchers presented six potential solutions. However, in late 2019 the RPA changed [80]. The change "was intended to accommodate and overcome claimed barriers to RPKI adoption", and distribution in machine-readable formats was approved. In the authors' opinion, although the change is in a good direction, the desired state of a good and simple configuration is still obstructed.

Low RPKI deployment penetration among CDNs

Content Delivery Networks (CDNs) have a crucial function in the global Internet, usually providing the user with large amounts of data in a fast, reliable, and inexpensive manner. Over half of all web traffic is estimated to be served over CDNs; therefore, securing this segment is a prominent concern globally.

Researchers [75] investigated some of the most prominent essential CDNs: Akamai, Amazon, CDNetworks, Chinacache, Cloudflare, Cotendo, Edgecast, Highwinds, Instart, Internap, Limelight, Mirrorimage,



(a) Usage of RPKI.

(b) Main reasons for not using RPKI.

Fig. 12. Results of a survey among NANOG and RIPE NCC communities.
Source: Data based on a survey by [9].

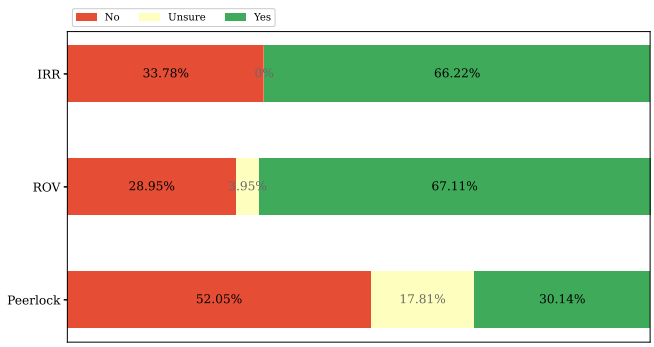


Fig. 13. Different countermeasures' usage among MANRS participants. Data based on a survey [74] of routing-conscious network administrators; therefore, values are higher than the global average.

Netdna, Simplecdn, and Yottaa. In this study, it is revealed that these CDNs consist of 199 different ASes. Of these, only four prefixes had an ROA issued, and these map to three origin ASes and one provider, Internap. These ASes are only a small fraction of the total, as Internap was operating 41 ASes. No other CDN has made any deployment.

To combat this low adoption process, in April 2020, MANRS launched a dedicated *program for CDN and Cloud Providers*. On December 3rd, 2020, the program had 14 members [81]. One of the mandatory actions is "fostering RPKI as the primary technology for validation of routing information on a global scale", while "improving consistency of route validation based on route objects published in an IRR".

8. Open issues and research challenges

The inter-domain security field presents multiple demands on any proposed control measure or concept. New proposals must be evaluated against several criteria regarding the theoretical, operational, and practical side of the problem. Future usage and adoption require acceptance from the research and industrial communities.

This section is dedicated to ongoing research directions and standards development paths, and it is divided into two parts. First, we discuss current open issues requiring closer attention. In the second part, we briefly review and give a sneak peek into the research challenges, including systems aiming to replace BGP and possible future Internet architectures.

8.1. Open issues

Next, we discuss the main technical obstacles to fully deployed and thriving inter-domain routing security. Then, we review the problems encountered and other feasibility studies, e.g., of RPKI usage.

RPKI sizing and scalability. There have been multiple concerns regarding the scalability, sizing, resiliency, and propagation times of the fully deployed RPKI system. Some attempts to approximate the size of the fully deployed RPKI system revealed that there is no consensus on the expected size of the global RPKI deployment, mainly due to different initial assumptions [3]. In a 2012 study, Osterweil et al. [82] envisioned 273k ROAs for 42k ASNs. In 2013, Kent et al. [83] predicted 289k ROAs for 43k ASNs; however, assuming that most ASes will not run an RPKI publication point but will use an RIR-provided repository, this number is actually significantly reduced. The 2013 study [84] estimated about 200k ROAs, when assuming aggregating three announcements in a single ROA.

Today in 2022, the advertised AS count is reaching 75,000. Assuming a pessimistic delegated RPKI model, where each AS hosts a repository and a RP validating software connects to each repository, 5,625 million connections are expected (75k * 75k). As shown in Fig. 9, the ROA count has surpassed 300k. Nevertheless, we struggle to find updated research on targeted RPKI deployment size; therefore, we treat this as an open question.

The related subject matter is the RPKI synchronization speed. This is the determinant of the time required to sync RP software with each PP repository. In the previously mentioned study [82], the estimated time for full synchronization without BGPsec is more than five days. Another piece of research [83] estimates around 60 min due to a smaller number of repositories.

The well-known longest possible synchronization time is crucial for a predictable propagation deadline of RPKI changes. Some use cases include DDoS attacks mitigation service providers [82] who use BGP advertisements to protect their clients, as well as network administrators executing defensive prefix deaggregation techniques or routine traffic engineering operations. Without ROV, these changes were possible within minutes. Therefore, in the best-case scenario, the propagation deadline would not impose an additional delay.

Rsync is the protocol commonly used by RPs for obtaining RPKI objects from PPs. However, [85] showed that usage of rsync is inefficient and insecure, e.g., a server is not protected from memory exhaustion. The new *RPKI Repository Delta Protocol* (RRDP) [86] was developed to combat these deficiencies. It was introduced just before 2020, and a

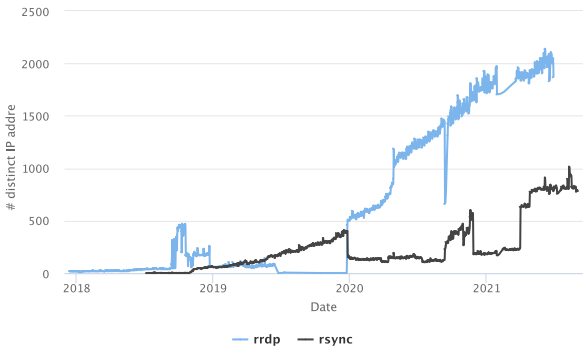


Fig. 14. RP activity by synchronization protocol.
Source: [88].

constant rise in RRDP popularity can be seen in Fig. 14. There is an ongoing effort to deprecate rsync for RPKI-related usage [87]. However, to achieve this goal, all RPs and PPs must first support RRDP, and then the phasing of rsync can begin.

A recent study [88] has checked a *timeliness*, i.e., what the current RPKI view is that RPs are seeing. It is a delicate balance between ensuring up-to-date data and over-straining the architecture or resources. On the other hand, an incomplete, stale, or outdated RPKI could lead to wrong routing decisions and cause a loss of reachability. Researchers using (i) controlled CAs and PPs, (ii) controlled RP, and (iii) controlled RPKI objects were able to gather statistics on the RPKI system usage. RPs using the newer RRDP protocol were more predictable than rsync. The connection interval varies for a particular implementation.

Currently, with the updated transfer protocol and RPs polling PP every 10 min to 60 min, the scaling issues seem considerable 1u [22, Section 4.1]. Nevertheless, the ROA and RPs count is on the constant rise; hence, we expect increased performance requirements for the RPKI system. The previously mentioned propagation deadlines are unlikely to be achieved at this moment.

RPKI maxLength parameter. The Route Origin Authorization record consists of the maxLength parameter. This specifies the maximal possible length of a subnet mask, creating the possibility for an ISP to cover multiple more specific IP prefixes under a single ROA. For example, an ROA for prefix 10.0.0.0/8 with maxLength /24 and a particular ASN states that this ASN can validly originate any prefix in the range of /8-/24, such as 10.0.0.0/16. This flexibility gives network operators a way of responding to a DDoS attack or executing traffic load-balancing.

However, setting the maxLength to longer than the originating prefix opens up the possibility of a forged origin sub-prefix hijack [89]. When an adversary announces the same prefix, with a length in the range of the maxLength and the original ASN, while the adverse ASN is in the AS-Path (a Type-N sub-prefix Hijack), the adversary attracts 100% of the traffic. Even worse, the hijack cannot be stopped by ROV, as the injected path is valid.

As of June 1, 2017, just 12% of prefixes in the RPKI used maxLength. Almost all of these prefixes (84%) are in ROAs that are not minimal and, thus, are vulnerable to forged-origin sub-prefix hijacks. Today, as the ROA count has grown, we expect more prefixes to be vulnerable.

In 2020, the ICANN – the non-profit organization that oversees the Internet – in technical analysis [22, Section 4.2] raised a point on the issue. There is also an IETF's BCP [90] recommendation released in 2022. Both documents agree that the only defense is issuing only minimal set of ROAs without the maxLength and only for prefixes that are currently announced. There are highlighted cases when urgent operational needs force issuing of a wider ROA. Thus, they are possible trade-offs between operational agility and security. With future path validation mechanism deployment, the issue is expected to become obsolete.

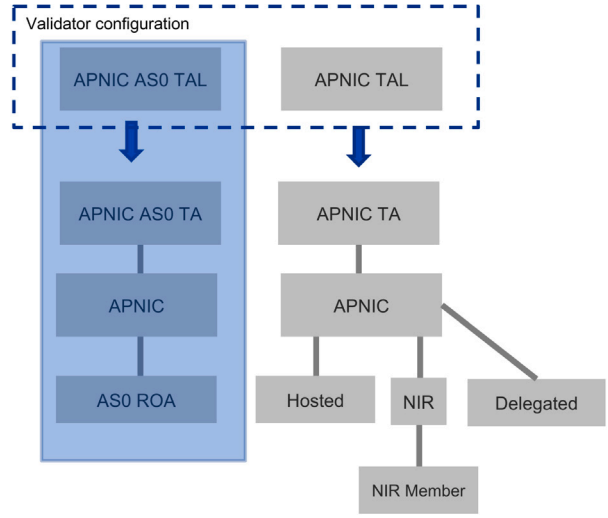


Fig. 15. The RPKI TA structure. AS0 is separate from the "main" TA; therefore, this TA can be used by opting in.
Source: [96]

Misbehaving RPKI authorities. Some attention has been brought to the excessive power of the RPKI authorities, i.e., RIRs. As they run and manage the private keys for RPKI's TAs, a singular authority can revoke an ROA or execute an IP prefix takedown. In our opinion, it is unlikely the RIR will undertake such an action on its own, but it could occur when coerced by the state, an LEA, or a court order. As always, malfunction or misconfiguration cannot be ruled out.

Researchers [91] introduced a new concept to the threat landscape by pointing out that authorities' power to overwrite a certificate additionally hinders detection, making it indistinguishable from regular RPKI activity. In [92], a recommendation for including mutual consent for prefix revocation is discussed. A proposal for limiting the power of a singular RIR has been mentioned [93] that can be achieved in the *Distributed RPKI* architecture, which utilizes the Multiparty Computation (MPC) with a threshold signatures algorithm.

AS0 ROA policy. The network community is not unanimous in handling unallocated IP address space routing. These prefixes are sometimes used in harmful ways, such as launching DDoS attacks or sending spam [94]. Unallocated prefixes are not meant to be used on the public Internet; therefore, a policy defining the standardized way of dealing with unassigned IP prefixes has been drafted. It uses ROAs with the origin of AS0 (the null AS), which is "an attestation by the holder of a prefix that the prefix described in the ROA, and any more specific prefix, should not be used in a routing context" [95].

Proponents of this solution emphasize the policy regulations—the AS0 ROA can be issued only by the resource owner and should only be possible for unallocated resources. It also has a lower priority than any ROA with a valid origin AS, so allocated resources are unaffected. The reason against [97] deployment is the abuse potential for network blocking and censorship. However, the policy states that AS0 ROA is possible only for unallocated resources.

This issue has triggered a discussion about whether the AS0 ROA record repository should be used by default (opt-out), operator consent (opt-in), or even at all. So far, only APNIC has implemented the policy in the separate TA, as seen in Fig. 15. This is independently validated under a different policy. The current recommendation is only for diagnostics and alerting usage, not to be applied in the BGP router calculations. AFRINIC and LACNIC have accepted the policy but have yet to deploy it. ARIN has no active policy discussion, and RIPE explicitly rejected the operation of an AS0 system.

8.2. Research challenges

There is an active interest in researching alternative ways to protect routing security. In the remainder of this section, we briefly discuss emerging and forward-looking approaches.

Software-Defined Networking (SDN). Some efforts are being put into evolving or even replacing the BGP with a new inter-domain routing protocol. With multiple proposals in opposition to the traditional distributed protocol, the Software Defined Network architecture stands out. This architecture decouples forwarding and control functions and simultaneously provides the ability to program the centrally managed network controller directly.

A summary of this work [98] concludes with a skeptical statement: "There is still much to be done until the BGP control plane is fully replaced as the main routing core system of the Internet. None of the efforts investigated in this work provides all the features to reach that goal." This valuable statement reveals that the SDN-based approach will not succeed globally in the immediate future.

However, there is a proposal [99] for a local application at the Internet eXchange Point (IXP). This enhances management capability by applying flexible software control methods, becoming the Software-Defined eXchange (SDX) center. The SD-BROV (SDX-based BGP Route Origin Validation) is a mechanism that provides a flexible route exchange scenario with RPKI validation. The validating application built into the SDN controller is capable of investigating the received routing information. Researchers claim that SD-BROV can detect and protect BGP incidents as long as a poisoned AS is connected to the SDX. Improvement of this shortcoming is on the future agenda.

Utilization of machine learning for anomaly detection. BGP anomaly detection systems conventionally use a time-series analysis of control plane events or reachability checks of the data plane. However, detection techniques have evolved to utilize Machine Learning (ML) approaches, as they can improve accuracy and identify a broader range of BGP anomalies.

A review [100] evaluated AS-level graph features through an extensive comparison of the different ML algorithms used in BGP anomaly detection, i.e., Naive Bayes (NB) classifier, Decision Trees (DT), Random Forests (RF), Support Vector Machines (SVM), Multi-Layer Perceptron (MLP), and Neural Networks (NN), which use graph features to detect BGP path leaks. According to the researchers, MLP achieved the highest accuracy; however, considering the other benefits of SVM, the best classifier is achieved using SVM.

The research community [101–103] seems to be interested in anomalies such as power outage events and worm propagation, but there is also a growing interest in detecting route leaks. Therefore, we anticipate more research to focus on improving the accuracy and quality of routing incident detection.

Recent advances include employing neural networks and unsupervised learning to detect routing attacks. These methods use Multi-Scale Long Short-Term Memory (MSLSTM) [104], AP2Vec—the adapted Word2Vec method for BGP [105,106], or the Graph Attention Network (GAT) [107].

Blockchain-based routing security. Mastilak et al. surveyed [108] the blockchain network's usage in routing security. Researchers evaluated eleven papers regarding the improvement they bring to inter-domain routing security; their performance, throughput, timeliness, and scalability are typically contradictory. The feature that most differentiates these projects from each other is the type of consensus algorithm. Typical approaches include *Proof of Work (PoW)*, *Proof of Stake (PoS)*, *Delegated Proof of Stake (DPoS)*, *Practical Byzantine Fault Tolerance (PBFT)*, *Proof of Authority (PoA)*, and *RAFT*. Thou, ROAChain [109] uses *Sharding* and RouteChain [110] uses *Clique*.

Those alternative approaches advocate for a decentralized ledger for managing Internet resources, which may prevent the abuse of the central authorities. The blockchain can be used as immutable storage for ROA records, IP block allocations, etc., with the advantage of high availability, as each node keeps a copy of the whole chain. Smart contracts have the potential to provide transparent business agreements or policies directly between peers in inter-domain routing.

The vast majority of blockchain-based routing projects are designed to coexist with existing countermeasures as a method to provide an additional layer of security. Although they can ensure higher transparency and proof of asset verification, the main drawbacks are scalability, throughput, and computing overhead. There is also an extensive list of possible attacks on the main chain or on a single network node. Blockchain-based routing security is an interesting concept; however, researchers [108] claims that considering its limitations and susceptibility to new attack surfaces, it is not production-ready.

Unlike traditional cryptocurrency blockchains, the Inter-domain Prefix and Route Validation (IPRV) framework [111] uses *Fast and Scalable DAG (FSD2L)*, a variant of the *Directed Acyclic Graph-based Distributed Ledger (DAG-DL)*, which is typically used for systems where a large number of transactions are continuously generated. The FSD2L allows for multi-signature transactions, which are critical for inter-domain routing; in this way, it prevents inconsistent states between ASes. The deployment of this solution requires each AS and RIR to operate a *Route Validation Node (RVN)*, which uses the *Proof of Authentication (PoA)* consensus. The RVN adds transactions (IP prefix allocations and BGP route changes) to the FSD2L while simultaneously allowing BGP speakers to determine the validity of the routes they receive from their peers. The authors claim that IPRV achieves high throughput thanks to scalability, low time to finality, and reduced storage requirements, while offering benefits that RPKI and BGPsec duo do not provide (e.g., addressing the problems we described in the section on Misbehaving RPKI Authorities). However, a standardized connector between routers and RVNs has yet to be developed.

SCION. There are also attempts [112] such as SCION, to design the so-called Future Internet Architectures—the entirely new, clean-slate architecture that aims to replace existing protocols and fix their problems. SCION claims it is not vulnerable to hijacking, Denial-of-Service, or spoofing attacks. In parallel, it provides some additional functionalities that are not security-related.

The biggest issue is the transition process, which cannot be performed overnight. However, SCION has been deployed in production by multiple academic, government, and financial industry entities, as it supports incremental deployability by the SCION-IP-Gateway (SIG). It has recently been presented to the IETF community [113].

The European Commission's flagship Next Generation Internet (NGI)[11] initiative is seeking a more democratic, inclusive, resilient, sustainable, and trustworthy Internet by 2030. The final report [114] covers several angles and needs; however, cybersecurity resilience is chosen as a goal in achieving the vision:

> In our vision for 2030, we will have made substantial strategic investments to improve the Internet's physical infrastructures by updating outdated systems, and will have designed and deployed systems that could withstand these new types of pressure and risk.

One of the supported projects is Verified SCION. By formally verifying SCION, it will be possible to produce proof with a guarantee of SCION security. This may accelerate BGP obsolescence.

[11] https://www.ngi.eu/ngi-projects/ngi-forward/

9. Conclusions

This work aimed to provide a comprehensive and up-to-date review of the routing security domain. The selected literature allowed us to gain theoretical knowledge about the field and synthesize the most-suited taxonomy of incidents; however, the addition of gray literature adds insights into the newest and most cutting-edge techniques and standards in development.

Based on the conducted review, we may formulate the following key findings, which are summarized here.

Incidents may originate from an action executed with malicious intent (service disruption, traffic interception, censorship, credential theft) or be accidentally caused by human error or misconfiguration. The overarching problem is the lack of BGP's native relation-aware security control measures being capable of validating peers' announcements in terms of intended scope, prefix propagation, and automated policy compliance verification.

Every routing anomaly is different. In addition to basic characteristics (prefix, AS-Path), we propose using the five-criteria benchmark (intent, scale, impact, duration, range). Thanks to this wide range of characteristics, it is possible to identify the essential features of each event.

Two existing leading taxonomies, hijacks and leaks, strongly overlap from a routing table standpoint. With unknown incentives and root causes, an incident can fit both taxonomies. We prefer using the one proposed in [6], as it covers more cases and is better logically structured and reflected in ARTEMIS software. However, the downside of this taxonomy is the lack of AS relationship awareness, a mandatory feature in the route leak domain.

Each type of incident requires a different approach. To secure against an origin hijack (the Type-0 attack class), the RPKI ROV is best-suited. However, until universal adoption is achieved by rejecting unknown prefixes, IRR filtering should be implemented in parallel. Path hijacks (the Type-N attack class) are the trickiest threats because only full path validation can ensure path correctness, and the only solution capable of doing path validation—BGPsec, is considered infeasible to deploy. Nevertheless, in the shadow lost opportunity of the BGPsec deployment, ASPA is coming to the forefront because of its right balance in deployment ease and security. Route leaks are receiving a lot of attention nowadays, with multiple proposals on the table. However, until the leading standard is chosen and its implementation is available, it is best to stick to IRR filtering, prefix count limit, and Peerlock implementation.

Incident response is challenging since incidents occur outside the victim network. Defensive techniques such as prefix deaggregation, the shortening of an AS-Path, and the withdrawal of an offending route are explained in Section 6. Due to limited options, organizations should prepare their incident response plan in advance.

To conclude, no universal tool or technique alone can fix or address all BGP vulnerabilities. In the final state, some combination of countermeasures likely needs to be deployed. The ongoing efforts to deploy the RPKI ROV are a significant step in improving routing security, but protecting the AS-Path is another endeavor. Hopefully, the growing interest in RPKI will create popularity and awareness of routing security.

Due to the non-existent central administration of the Internet, the future of routing security is in the hands of the network operators' community. Fixing the deficits requires collective action; therefore, we can only ensure a more secure Internet as a cooperative community.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mikolaj Kowalski reports a relationship with Orange Poland that includes: employment.

Data availability

No data was used for the research described in the article.

References

[1] J.A. Hawkinson, T.J. Bates, Guidelines for creation, selection, and registration of an Autonomous System (AS), RFC 1930, 1996, http://dx.doi.org/10.17487/RFC1930, URL https://rfc-editor.org/rfc/rfc1930.txt.

[2] A. Mitseva, A. Panchenko, T. Engel, The state of affairs in BGP security: A survey of attacks and defenses, Comput. Commun. 124 (2018) 45–60, http://dx.doi.org/10.1016/j.comcom.2018.04.013.

[3] M.S. Siddiqui, D. Montero, R. Serral-Gracià, X. Masip-Bruin, M. Yannuzzi, A survey on the recent efforts of the Internet Standardization Body for securing inter-domain routing, Comput. Netw. 80 (2015) 1–26, http://dx.doi.org/10.1016/j.comnet.2015.01.017, URL https://www.sciencedirect.com/science/article/pii/S1389128615000286.

[4] B. Al-Musawi, P. Branch, G. Armitage, BGP anomaly detection techniques: A survey, IEEE Commun. Surv. Tutorials 19 (1) (2017) 377–396, http://dx.doi.org/10.1109/comst.2016.2622240.

[5] K. Sriram, D. Montgomery, D.R. McPherson, E. Osterweil, B. Dickson, Problem definition and classification of BGP route leaks, RFC 7908, 2016, http://dx.doi.org/10.17487/RFC7908, URL https://rfc-editor.org/rfc/rfc7908.txt.

[6] P. Sermpezis, V. Kotronis, P. Gigis, X. Dimitropoulos, D. Cicalese, A. King, A. Dainotti, ARTEMIS: Neutralizing BGP hijacking within a minute, IEEE/ACM Trans. Netw. 26 (6) (2018) 2471–2486, http://dx.doi.org/10.1109/TNET.2018.2869798, arXiv:1801.01085.

[7] G. Huston, A survey on securing inter-domain routing, 2021, The ISP Column, URL https://www.potaroo.net/ispcol/2021-07/sidr1.pdf.

[8] J. Durand, I. Pepelnjak, G. Doering, BGP operations and security, RFC 7454, 2015, http://dx.doi.org/10.17487/RFC7454, URL https://rfc-editor.org/rfc/rfc7454.txt.

[9] P. Sermpezis, V. Kotronis, A. Dainotti, X. Dimitropoulos, A survey among network operators on BGP prefix hijacking, ACM SIGCOMM Comput. Commun. Rev. 48 (1) (2018) 64–69, http://dx.doi.org/10.1145/3211852.3211862, arXiv:1801.02918.

[10] Y. Rekhter, S. Hares, T. Li, A border gateway protocol 4 (BGP-4), RFC 4271, 2006, http://dx.doi.org/10.17487/RFC4271, URL https://www.rfc-editor.org/info/rfc4271.

[11] C.E. Perkins, J.A. Bharatia, P.R. Calhoun, Mobile IPv4 challenge/response extensions (revised), RFC 4721, 2007, http://dx.doi.org/10.17487/RFC4721, URL https://www.rfc-editor.org/info/rfc4721.

[12] Catchpoint, The comprehensive guide to BGP, 2021, URL https://www.bgp.org/blog/x-raying-bgp.

[13] L. Gao, On inferring autonomous system relationships in the Internet, IEEE/ACM Trans. Netw. 9 (6) (2001) 733–745, http://dx.doi.org/10.1109/90.974527.

[14] L. Gao, J. Rexford, Stable Internet routing without global coordination, IEEE/ACM Trans. Netw. 9 (6) (2001) 681–692, http://dx.doi.org/10.1109/90.974523.

[15] X. Dimitropoulos, D. Krioukov, M. Fomenkov, B. Huffaker, Y. Hyun, k. claffy, G. Riley, AS relationships: Inference and validation, ACM SIGCOMM Comput. Commun. Rev. (CCR) 37 (2007) 29–40, http://dx.doi.org/10.1145/1198255.1198259.

[16] IANA, Number Resources, URL https://www.iana.org/numbers.

[17] S.L. Murphy, BGP security vulnerabilities analysis, RFC 4272, 2006, http://dx.doi.org/10.17487/RFC4272, URL https://www.rfc-editor.org/info/rfc4272.

[18] A. Heffernan, Protection of BGP sessions via the TCP MD5 signature option, RFC 2385, 1998, http://dx.doi.org/10.17487/RFC2385, URL https://www.rfc-editor.org/info/rfc2385.

[19] D.J.D. Touch, R. Bonica, A.J. Mankin, The TCP authentication option, RFC 5925, 2010, http://dx.doi.org/10.17487/RFC5925, URL https://www.rfc-editor.org/info/rfc5925.

[20] C. Pignataro, P. Savola, D. Meyer, V. Gill, J. Heasley, The generalized TTL security mechanism (GTSM), RFC 5082, 2007, http://dx.doi.org/10.17487/RFC5082, URL https://www.rfc-editor.org/info/rfc5082.

[21] K. Seo, S. Kent, Security architecture for the internet protocol, RFC 4301, 2005, http://dx.doi.org/10.17487/RFC4301, URL https://www.rfc-editor.org/info/rfc4301.

[22] A. Durand, Resource public key infrastructure (RPKI) technical analysis, techreport, ICANN Office of the Chief Technology Officer, 2020, URL https://www.icann.org/en/system/files/files/octo-014-02sep20-en.pdf.

[23] P. Boothe, J. Hiebert, R. Bush, Short-lived prefix hijacking on the internet, in: Proc. of the NANOG 36, 2006, URL https://archive.nanog.org/meetings/nanog36/presentations/boothe.pdf.

[24] T. Hassan, P. Bjorke, The Hunt for 3ve, Tech. rep., White Ops, Google, 2018, URL https://services.google.com/fh/files/blogs/3ve_google_whiteops_whitepaper_final_nov_2018.pdf.

[25] T. Dai, P. Jeitner, H. Shulman, M. Waidner, The hijackers guide to the galaxy: Off-path taking over internet resources, in: 30th USENIX Security Symposium (USENIX Security 21), USENIX Association, 2021, pp. 3147–3164, URL https://www.usenix.org/conference/usenixsecurity21/presentation/dai.

[26] T. Dai, P. Jeitner, H. Shulman, M. Waidner, From IP to transport and beyond: cross-layer attacks against applications, in: Proceedings of the 2021 ACM SIGCOMM 2021 Conference, ACM, 2021, pp. 836–849, http://dx.doi.org/10.1145/3452296.3472933, arXiv:2205.06085.

[27] T. Hlavacek, P. Jeitner, D. Mirdita, H. Shulman, M. Waidner, Behind the scenes of RPKI, in: Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security, ACM, 2022, http://dx.doi.org/10.1145/3548606.3560645.

[28] T. Hlavacek, P. Jeitner, D. Mirdita, H. Shulman, M. Waidner, Stalloris: RPKI downgrade attack, in: 31st USENIX Security Symposium (USENIX Security 22), USENIX Association, Boston, MA, 2022, pp. 4455–4471, URL https://www.usenix.org/conference/usenixsecurity22/presentation/hlavacek.

[29] A. Gavrichenkov, Breaking HTTPS with BGP hijacking, in: Black Hat USA 2015, 2015, URL https://www.blackhat.com/us-15/briefings.html#breaking-https-with-bgp-hijacking.

[30] H. Birge-Lee, Y. Sun, A. Edmundson, J. Rexford, P. Mittal, Using BGP to acquire bogus TLS certificates, in: The 17th Privacy Enhancing Technologies Symposium, Princeton Univeristy, 2017, URL https://www.princeton.edu/~pmittal/publications/bgp-tls-hotpets17.

[31] H. Birge-Lee, Y. Sun, A. Edmundson, J. Rexford, P. Mittal, Bamboozling certificate authorities with BGP, in: Proceedings of the 27th USENIX Conference on Security Symposium, SEC '18, USENIX Association, 2018, pp. 833–849, URL https://www.cs.princeton.edu/~jrex/papers/bamboozle18.pdf.

[32] A. Gavrichenkov, Four Years of Breaking HTTPS with BGP Hijacking, Qrator Labs CZ, 2019, URL https://pc.nanog.org/static/published/meetings//NANOG75/daily/day_3.html#talk_1892.

[33] R. Brandom, Hackers emptied Ethereum wallets by breaking the basic infrastructure of the internet, 2018, URL https://www.theverge.com/2018/4/24/17275982/myetherwallet-hack-bgp-dns-hijacking-stolen-ethereum.

[34] A. Toonk, Massive route leak causes Internet slowdown, 2015, URL https://www.bgpmon.net/massive-route-leak-cause-internet-slowdown/.

[35] G. Huston, More leaky routes, 2015, RIPE Labs, URL https://labs.ripe.net/author/gih/more-leaky-routes/.

[36] N. Kephart, Route leak causes amazon and AWS outage, 2015, URL https://www.thousandeyes.com/blog/route-leak-causes-amazon-and-aws-outage.

[37] M. d'Itri, Hacking Team and a case of BGP hijacking, 2015, URL http://blog.bofh.it/id_456.

[38] Hacking Team, Anonymizer ROS, 2013, Wikileaks, URL https://www.wikileaks.org/hackingteam/emails/emailid/347856.

[39] D. Madory, M. Prosser, Excessive BGP AS path prepending is a self-inflicted vulnerability, in: RIPE 79, Oracle Internet Intelligence, 2019, URL https://ripe79.ripe.net/presentations/64-prepending_madory2.pdf.

[40] D. Kong, R. Watro, K. Seo, S. Kent, Certificate policy (CP) for the resource public key infrastructure (RPKI), RFC 6484, 2012, http://dx.doi.org/10.17487/RFC6484, URL https://www.rfc-editor.org/info/rfc6484.

[41] R. Housley, Cryptographic message syntax (CMS), RFC 5652, 2009, http://dx.doi.org/10.17487/RFC5652, URL https://www.rfc-editor.org/info/rfc5652.

[42] P. Mohapatra, J. Scudder, D. Ward, R. Bush, R. Austein, BGP prefix origin validation, RFC 6811, 2013, http://dx.doi.org/10.17487/RFC6811, URL https://www.rfc-editor.org/info/rfc6811.

[43] R. Bush, R. Austein, The resource public key infrastructure (RPKI) to router protocol, version 1, RFC 8210, 2017, http://dx.doi.org/10.17487/RFC8210, URL https://www.rfc-editor.org/info/rfc8210.

[44] M. Lepinski, K. Sriram, Bgpsec protocol specification, RFC 8205, 2017, http://dx.doi.org/10.17487/RFC8205, URL https://rfc-editor.org/rfc/rfc8205.txt.

[45] S.T. Kent, C. Lynn, J. Mikkelson, K. Seo, Secure border gateway protocol (s-BGP) - real world performance and deployment issues, in: Network and Distributed System Security Symposium NDSS 2000, 2000, URL https://www.ndss-symposium.org/wp-content/uploads/2017/09/Secure-Border-Gateway-Protocol-S-BGP-Real-World-Performance-and-Deployment-Issues-paperStephen-Kent.pdf.

[46] R. Lychev, S. Goldberg, M. Schapira, BGP security in partial deployment: Is the juice worth the squeeze? 2013, CoRR arXiv:1307.2690.

[47] G. Goodell, W. Aiello, T. Griffin, J. Ioannidis, P. Mcdaniel, A. Rubin, Working around BGP: An incremental approach to improving security and accuracy of interdomain routing, in: Proceedings of the Network and Distributed System Security Symposium, NDSS 2003, San Diego, California, USA, 2002.

[48] T. Wan, E. Kranakis, P.C. van Oorschot, Pretty secure BGP, psBGP, in: Proceedings of the Network and Distributed System Security Symposium, NDSS 2005, San Diego, California, USA, The Internet Society, 2005, URL https://www.ndss-symposium.org/ndss2005/pretty-secure-bgp-psbgp/.

[49] R. White, Architecture and deployment considerations for secure origin BGP (soBGP), 2006, URL https://datatracker.ietf.org/doc/html/draft-white-sobgp-architecture-02, in preparation.

[50] W. George, S.L. Murphy, BGPsec considerations for autonomous system (AS) migration, RFC 8206, 2017, http://dx.doi.org/10.17487/RFC8206, URL https://rfc-editor.org/rfc/rfc8206.txt.

[51] O. Borchert, BGP secure routing extension (BGP-SRx) prototype, Tech. rep., NIST, 2016, URL https://www.nist.gov/services-resources/software/bgp-secure-routing-extension-bgp-srx-prototype, 2021-01-06.

[52] O. Junjie, N. Yanai, T. Takemura, M. Okada, S. Okamura, J.P. Cruz, APVAS: Reducing memory size of AS_PATH validation by using aggregate signatures, 2020, http://dx.doi.org/10.48550/arXiv.2008.13346, arXiv:2008.13346.

[53] S. Kent, A. Chi, Threat model for BGP path security, RFC 7132, 2014, http://dx.doi.org/10.17487/RFC7132, URL https://www.rfc-editor.org/info/rfc7132.

[54] D.R. McPherson, S. Amante, E. Osterweil, D. Mitchell, Route-leaks & MITM attacks against BGPSEC, 2014, URL https://datatracker.ietf.org/doc/draft-ietf-grow-simple-leak-attack-bgpsec-no-help/04/, in preparation.

[55] J. Mauch, J. Snijders, G. Hankins, Default external BGP (EBGP) route propagation behavior without policies, RFC 8212, 2017, http://dx.doi.org/10.17487/RFC8212, URL https://www.rfc-editor.org/info/rfc8212.

[56] J. Snijders, NTT peer locking, Tech. rep., NTT Global NOC, 2016, URL http://instituut.net/~job/peerlock_manual.pdf.

[57] T. McDaniel, J.M. Smith, M. Schuchard, Flexsealing BGP against route leaks: Peerlock active measurement and analysis, in: Proceedings 2021 Network and Distributed System Security Symposium, Internet Society, 2021, http://dx.doi.org/10.14722/ndss.2021.23080, arXiv:2006.06576.

[58] J. Snijders, Everyday practical BGP filtering, in: NANOG 67 Proceedings, 2016.

[59] A. Azimov, E. Uskov, R. Bush, K. Patel, J. Snijders, R. Housley, A profile for autonomous system provider authorization, 2022, URL https://datatracker.ietf.org/doc/html/draft-ietf-sidrops-aspa-profile-07, in preparation.

[60] A. Azimov, E. Bogomazov, R. Bush, K. Patel, J. Snijders, Verification of AS_PATH using the resource certificate public key infrastructure and autonomous system provider authorization, 2021, URL https://datatracker.ietf.org/doc/html/draft-ietf-sidrops-aspa-verification-08, in preparation.

[61] A. Azimov, E. Bogomazov, R. Bush, K. Patel, K. Sriram, Route leak prevention and detection using roles in UPDATE and open messages, RFC 9234, 2022, http://dx.doi.org/10.17487/RFC9234, URL https://www.rfc-editor.org/info/rfc9234.

[62] K. Sriram, A. Azimov, Methods for detection and mitigation of BGP route leaks, 2022, URL https://datatracker.ietf.org/doc/html/draft-ietf-grow-route-leak-detection-mitigation-07, in preparation.

[63] A. Cohen, Y. Gilad, A. Herzberg, M. Schapira, Jumpstarting BGP security with path-end validation, in: Proceedings of the 2016 ACM SIGCOMM Conference, ACM, 2016, http://dx.doi.org/10.1145/2934872.2934883.

[64] W. Brand, J. Posen, A reproduction of "jumpstarting BGP security with path-end validation", Tech. rep., Stanford CS244, 2020, URL https://reproducingnetworkresearch.files.wordpress.com/2020/06/brand_posen.pdf.

[65] T. Collyer, BGP hijinks and hijacks - incident response when your backbone is your enemy, Tech. rep., SANS, 2016, URL https://sansorg.egnyte.com/dl/VPuN2ZPrjI.

[66] F.V. Silveira, RPKI: Past, present and future, 2020, RIPE Labs, URL https://labs.ripe.net/author/felipe_victolla_silveira/rpki-past-present-and-future/.

[67] Cloudflare RPKI portal, 2022, URL https://rpki.cloudflare.com/.

[68] A. Reuter, R. Bush, I. Cunha, E. Katz-Bassett, T.C. Schmidt, M. Wählisch, Towards a rigorous methodology for measuring adoption of RPKI route validation and filtering, ACM SIGCOMM Comput. Commun. Rev. 48 (1) (2018) 19–27, http://dx.doi.org/10.1145/3211852.3211856, arXiv:1706.04263.

[69] NIST RPKI monitor, 2022, URL https://rpki-monitor.antd.nist.gov.

[70] B. Claise, Cisco systems NetFlow services export version 9, RFC 3954, 2004, http://dx.doi.org/10.17487/RFC3954, URL https://www.rfc-editor.org/info/rfc3954.

[71] J. Snijders, Looking ahead: BGP routing security, in: DKNOG9 Proceedings, 2019, URL https://events.dknog.dk/event/4/contributions/37/attachments/15/25/DKNOG9_Snijders_Routing_security_roadmap1.pdf.

[72] D. Madory, J. Snijders, Measuring RPKI ROV adoption with NetFlow, in: Proceedings of NANOG84, 2022, URL https://storage.googleapis.com/site-media-prod/meetings/NANOG84/2488/20220214_Madory_Measuring_Rpki_Rov_v1.pdf.

[73] Federal Communications Commission, Secure internet routing, 2022.

[74] J.L. Hall, O. Kolkman, M. Kruse, R. Polk, A. Robachevsky, A. Siddiqui, Comments of internet society, Tech. rep., The Internet Society (ISOC), 2022, PS Docket No. 22-90, URL https://www.internetsociety.org/wp-content/uploads/2022/04/ISOC-FCC-NOI-Comments-04112022.pdf.

[75] M. Wählisch, R. Schmidt, T.C. Schmidt, O. Maennel, S. Uhlig, G. Tyson, RiPKI: The tragic story of RPKI deployment in the web ecosystem, in: Proceedings of the 14th ACM Workshop on Hot Topics in Networks, ACM, 2015, http://dx.doi.org/10.1145/2834050.2834102, arXiv:1408.0391.

[76] T. Hlavacek, I. Cunha, Y. Gilad, A. Herzberg, E. Katz-Bassett, M. Schapira, H. Shulman, DISCO: Sidestepping RPKI's deployment barriers, in: Proceedings 2020 Network and Distributed System Security Symposium, Internet Society, 2020, http://dx.doi.org/10.14722/ndss.2020.24355.

[77] A. Robachevsky, Improving routing security through concerted action, in: Proceedings on RIPE 80, 2020, URL https://ripe80.ripe.net/wp-content/uploads/presentations/3-202005-MANRS-RIPE80.pdf.

[78] B. Cartwright-Cox, Are BGPs security features working yet?, 2018, URL https://blog.benjojo.co.uk/post/are-bgps-security-features-working-yet-rpki.

[79] C.S. Yoo, D. Wishnick, Lowering legal barriers to RPKI adoption, SSRN Electron. J. (19–02) (2019) http://dx.doi.org/10.2139/ssrn.3308619.
[80] J. Curran, ARIN announces new relying party agreement (RPA) to spur use of RPKI, 2019, URL https://www.arin.net/vault/announcements/2019/20191021.html.
[81] A. Siddiqui, MANRS welcomes 500th network operator, 2020, URL https://www.manrs.org/2020/12/manrs-welcomes-500th-network-operator/.
[82] E. Osterweil, T. Manderson, Sizing estimates for a fully deployed RPKI, Tech. rep., Verisign Labs, 2012.
[83] S. Kent, K. Sriram, RPKI rsync download delay modeling, in: IETF-86 SIDR WG Meeting, 2013.
[84] T. Bruijnzeels, O. Muravskiy, B. Weber, RPKI repository analysis and requirements, 2013, URL https://datatracker.ietf.org/doc/draft-tbruijnzeels-sidr-repo-analysis/00/, in preparation.
[85] G. Michaelson, B. Ellacot, Rsync considered inefficient and harmful, in: Proceedings of IETF 89, 2014.
[86] T. Bruijnzeels, O. Muravskiy, B. Weber, R. Austein, The RPKI Repository Delta protocol (RRDP), RFC 8182, 2017, http://dx.doi.org/10.17487/RFC8182, URL https://www.rfc-editor.org/info/rfc8182.
[87] T. Bruijnzeels, R. Bush, G.G. Michaelson, Resource public key infrastructure (RPKI) repository requirements, 2021, URL https://datatracker.ietf.org/doc/draft-ietf-sidrops-prefer-rrdp/01/, in preparation.
[88] J. Kristoff, R. Bush, C. Kanich, G. Michaelson, A. Phokeer, T.C. Schmidt, M. Wählisch, On measuring RPKI relying parties, in: Proceedings of the ACM Internet Measurement Conference, ACM, 2020, http://dx.doi.org/10.1145/3419394.3423622.
[89] Y. Gilad, O. Sagga, S. Goldberg, MaxLength considered harmful to the RPKI, in: Proceedings of the 13th International Conference on Emerging Networking EXperiments and Technologies, CoNEXT '17, Association for Computing Machinery, New York, NY, USA, 2017, pp. 101–107, http://dx.doi.org/10.1145/3143361.3143363.
[90] Y. Gilad, S. Goldberg, K. Sriram, J. Snijders, B. Maddison, The use of maxlength in the resource public key infrastructure (RPKI), RFC 9319, 2022, http://dx.doi.org/10.17487/RFC9319, URL https://www.rfc-editor.org/info/rfc9319.
[91] D. Cooper, E. Heilman, K. Brogle, L. Reyzin, S. Goldberg, On the risk of misbehaving RPKI authorities, in: Proceedings of the Twelfth ACM Workshop on Hot Topics in Networks, ACM, 2013, http://dx.doi.org/10.1145/2535771.2535787.
[92] E. Heilman, D. Cooper, L. Reyzin, S. Goldberg, From the consent of the routed, in: Proceedings of the 2014 ACM Conference on SIGCOMM, ACM, 2014, http://dx.doi.org/10.1145/2619239.2626293.
[93] K. Shrishak, H. Shulman, Limiting the power of RPKI authorities, in: Proceedings of the Applied Networking Research Workshop, ACM, 2020, http://dx.doi.org/10.1145/3404868.3406674, URL https://irtf.org/anrw/2020/slides-shrishak-00.pdf.
[94] A.L. Mathurin, What's the AS0 ROA policy, and what should I know as a network operator?, 2020, URL https://www.manrs.org/2020/12/whats-the-as0-roa-policy-and-what-should-i-know-as-a-network-operator/.
[95] G. Huston, G.G. Michaelson, Validation of route origination using the resource certificate public key infrastructure (PKI) and route origin authorizations (ROAs), RFC 6483, 2012, http://dx.doi.org/10.17487/RFC6483, URL https://www.rfc-editor.org/info/rfc6483.
[96] G. Michaelson, Demystifying AS0, in: Proceeding APNIC52, 2021.
[97] J. Snijders, The dangers of demonizing bogons 2019-08, 2020, Mailinglist routing-wg, URL https://www.ripe.net/ripe/mail/archives/routing-wg/2020-June/004131.html.
[98] W. Silva, D. Sadok, A survey on efforts to evolve the control plane of inter-domain routing, Information 9 (5) (2018) 125, http://dx.doi.org/10.3390/info9050125.
[99] P.-W. Tsai, A. Risdianto, M. Choi, S. Permal, T.C. Ling, SD-BROV: An enhanced BGP hijacking protection with route validation in software-defined exchange, Future Internet 13 (2021) 171, http://dx.doi.org/10.3390/fi13070171.
[100] O.R. Sanchez, S. Ferlin, C. Pelsser, R. Bush, Comparing machine learning algorithms for BGP anomaly detection using graph features, in: Proceedings of the 3rd ACM CoNEXT Workshop on Big DAta, Machine Learning and Artificial Intelligence for Data Communication Networks, in: Big-DAMA '19, ACM, 2019, pp. 35–41, http://dx.doi.org/10.1145/3359992.3366640.
[101] M. Cosovic, S. Obradovic, E. Junuz, Deep learning for detection of BGP anomalies, in: Time Series Analysis and Forecasting, Springer International Publishing, 2018, pp. 95–113, http://dx.doi.org/10.1007/978-3-319-96944-2_7.
[102] A. Bhatnagar, N. Majumdar, S. Shukla, BGP anomaly detection using decision tree based machine learning classifiers, Int. J. Innov. Technol. Explor. Eng. 8 (12) (2019) 4015–4020, http://dx.doi.org/10.35940/ijitee.l3622.1081219.
[103] X. Dai, N. Wang, W. Wang, Application of machine learning in BGP anomaly detection, J. Phys. Conf. Ser. 1176 (2019) 032015, http://dx.doi.org/10.1088/1742-6596/1176/3/032015.
[104] M. Cheng, Q. Li, J. Lv, W. Liu, J. Wang, Multi-scale LSTM model for BGP anomaly classification, IEEE Trans. Serv. Comput. 14 (3) (2021) 765–778, http://dx.doi.org/10.1109/tsc.2018.2824809.
[105] T. Shapira, Y. Shavitt, A deep learning approach for IP hijack detection based on ASN embedding, in: Proceedings of the Workshop on Network Meets AI & ML, ACM, 2020, http://dx.doi.org/10.1145/3405671.3405814.
[106] T. Shapira, Y. Shavitt, AP2Vec: an unsupervised approach for BGP hijacking detection, IEEE Trans. Netw. Serv. Manag. (2022) 1, http://dx.doi.org/10.1109/tnsm.2022.3166450.
[107] S. Peng, J. Nie, X. Shu, Z. Ruan, L. Wang, Y. Sheng, Q. Xuan, A multi-view framework for BGP anomaly detection via graph attention network, Comput. Netw. 214 (2022) 109129, http://dx.doi.org/10.1016/j.comnet.2022.109129, arXiv:2112.12793.
[108] L. Mastilak, P. Helebrandt, M. Galinski, I. Kotuliak, Secure inter-domain routing based on blockchain: A comprehensive survey, Sensors 22 (4) (2022) 1437, http://dx.doi.org/10.3390/s22041437.
[109] G. He, W. Su, S. Gao, J. Yue, S.K. Das, ROAchain: Securing route origin authorization with blockchain for inter-domain routing, IEEE Trans. Netw. Serv. Manag. 18 (2) (2021) 1690–1705, http://dx.doi.org/10.1109/tnsm.2020.3015557.
[110] M. Saad, A. Anwar, A. Ahmad, H. Alasmary, M. Yuksel, D. Mohaisen, RouteChain: Towards blockchain-based secure and efficient BGP routing, Comput. Netw. 217 (2022) 109362, http://dx.doi.org/10.1016/j.comnet.2022.109362.
[111] P. Podili, S.R. Cherupally, S. Boga, K. Kataoka, Inter-domain prefix and route validation using fast and scalable DAG based distributed ledger for secure BGP routing, J. Netw. Syst. Manage. 30 (4) (2022) http://dx.doi.org/10.1007/s10922-022-09668-2.
[112] A. Perrig, P. Szalachowski, R.M. Reischuk, L. Chuat, in: R. Nugent (Ed.), SCION: A Secure Internet Architecture, Springer International Publishing, 2017, p. 452, http://dx.doi.org/10.1007/978-3-319-67080-5.
[113] C. de Kater, N. Rustignoli, A. Perrig, SCION overview, 2022, URL https://datatracker.ietf.org/doc/draft-dekater-panrg-scion-overview/01/, in preparation.
[114] K. Bego, M. Droemann, A vision for the future internet, Tech. rep., Next Generation Internet, 2020, URL https://research.ngi.eu/wp-content/uploads/2021/02/Vision-for-the-future-internet-long-version-final-1.pdf.

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